

6.1 Introduction

To begin the design of any structure, the designer must know the type of soil on which the structure will be built. So, the first step in design of an offshore structure platform is a soil investigation, but the offshore soil investigation technique is not like onshore soil investigation.

A fixed offshore platform is defined as a jacket-type structure extending above water level and supported by relatively long piles, because this type of structure has been an industry standard for many years. Of course, nowadays other types of structures are being built, many different designs are being proposed, and the soil investigation requirements and the piles' depth depends on the type of structure, since these factors could vary considerably for different types of structures.

For a typical pile-supported platform, the soil investigation usually consists of a single bored sampled to a penetration below the expected pile-tip elevation.

6.2 Geotechnical investigation

The final design of a fixed offshore platform is based on the best soil field test data that can be obtained at the exact structure location. Therefore, an accurate survey should be implemented with the soil investigation. In general, the soil data are usually needed well in advance of any construction at a site.

The marine vessel doing the soil investigation should have a mooring or positioning system and should be capable of drilling and sampling well below the maximum expected penetration of the piles. Furthermore, offshore soil investigation is characteristically more costly than similar work onshore, due to the cost of the marine vessel itself, as its fee is calculated per day, which is expensive. This is a main difference between soil investigation onshore and offshore soil investigation. Therefore, the objectives of an investigation should always be kept in mind and the planning effort and selection of equipment and techniques should be aimed at achieving the objectives at the lowest cost.

Once the desired field investigation has been completed, further investigation is needed in the laboratory to evaluate and soil properties in order to apply the field and laboratory results to the design problem. It is appropriate to mention at this point that a fairly wide range of results can be obtained, particularly in the field,

depending on several factors. It is essential that the factors influencing the results be evaluated in interpreting the data and applying the results to the design.

The observations made during drilling and sampling should be recorded, because they can also provide some indication of potential problems in pile installation.

6.2.1 *Performing an offshore investigation*

An investigation at sea must begin by identifying the location, which is already done by a survey boat. A typical vessel for soil investigation has a length in the range of 40 to 70 m. The vessel is equipped with winches, cables, and anchors for four-point mooring. Anchor lines usually approach about eight times water depth.

Pile foundations frequently penetrate 90–120 m into the sea bottom. The combination of water depth and bottom penetration means that drilling equipment for soil borings should have a depth capability of 300 m or more.

In areas of very soft, underconsolidated soils, it is necessary to exercise very careful control of drilling fluid weight to counteract the tendency of the material to squeeze into the drilled hole and up into the drill pipe. Failure to do so can result in very severe sample disturbance. Furthermore, in such areas, the problem of handling disturbance on recovered samples is also quite severe.

Once a boring has penetrated a granular soil formation, it is essential that drilling mud having suitable viscosity and gel properties, which will be used to stabilize the drilled hole and to prevent caving; commercial saltwater gel is excellent for this purpose. Particularly in glacial deposits, coarse granular material such as cobbles or boulders may be encountered and make drilling extremely difficult. The presence of rock formations within the depth of investigation requires that special tools and procedures be used.

In most cases, gas be present within formations penetrated by soil borings and flows of water may also be encountered. The normal procedure of using a blowout preventer on cased holes is not easily applied to the wire-line method; the only reasonable protection against blowout is through a controlled mud-weight program using commercial barites.

6.3 Soil tests

The wet rotary process commonly used in advance of onshore soil boring is used also for offshore soil testing. In this respect, there is little difference between onshore and offshore practice except in some details in the way the objective is accomplished. However, the different offshore environmental conditions have necessitated changes in soil sampling procedures. An understanding of onshore sampling techniques, tools, and results will aid in understanding the required alterations, the concessions made, and the advantages of the offshore procedures.

Due to the cost of soil investigation offshore, preliminary engineering is recommended. Many sources of useful information and data about soil characteristics can

be gained from the geologic information about the platform location, drilling records, and acoustic data.

Many in-situ testing devices have been developed and are used to determine soil properties and conditions in the ground. Among them are the cone penetrometer, commonly known as the *Dutch cone*; the pressure meter; and the vane shear device. None of these provides a sample for other tests; if samples are required, sampling must be done between the in-situ tests or in a companion boring.

The Dutch cone was developed principally to define granular soil strata that would serve to support point-bearing piles. The cone is modified to measure both side friction and point-bearing resistance. Experienced operators claim to be able to identify soil types by the ratios of these resistances. Modern cone equipment has been used successfully only in shallow penetrations at sea. Remotely operated sea floor equipment presently being developed in Europe may have a substantially greater depth capability.

The pressure meter is designed to determine soil behavior by using an expanding pressure cell to measure load-deformation characteristics. The pressure cell is either driven or pushed into undisturbed soil to begin testing or is installed in the bottom of a drilled hole of carefully controlled dimensions. Readings, interpreted in terms of modulus of deformation and limit pressure, are used to determine bearing capacity, settlement, and other data. This tool has seen little success in connection with deep exploration for pile foundations at sea.

The remote vane, which is the vane shear device, has been used for years to measure in-place shear strengths of soils; however, soil shear failure was produced through torque applied to rods extending to the surface.

Remotely operated vane equipment has been developed in recent years and has been used successfully in a number of offshore investigations.

Tests can be conducted fairly quickly and economically in materials having a shear strength up to about 192 kN/m^2 . The present vane probe is powered electrically through a conductor cable extending to the surface but has no connection to the drill pipe. Continuous torque readings throughout a test are monitored on a readout at the surface. Development is under way on another device that has no electrical connection to the surface; readings will be stored in the down-hole device and will be read when it is brought to the surface after a test.

Results of the remote vane tests show strengths consistently higher than can be measured on samples recovered. This is to be expected because of disturbance created by sampling and sample handling and because of the pressure relief experienced by samples brought to the surface. The vane shear device probably better measures the true in-situ shear strength of a cohesive soil than any other device.

Until the principal emphasis shifts from pile-supported platforms, which is not expected any time soon, and until offshore activity extends into much deeper water, the present methods and techniques of offshore soil investigation will continue to be employed. Boat drilling and wire-line sampling methods offer the most versatile and economical means of investigation. Use of remote devices, such as the vane probe and cone penetrometer, in drilled holes, will increase and add much to our knowledge of in-situ soil properties. Dynamic positioning of surface drilling vessels

or support vessels for other operations will become more important as operations move into deeper waters beyond the continental shelf. There will be much more activity in the development of remotely operated sea floor drilling and sampling devices. Manned submersibles should also play an increasingly important role with further development.

6.4 In-situ testing

As mentioned previously, offshore soil investigation is much more expensive than onshore investigation. Therefore, the Nosrok (2004) report recommends that all in-situ test equipment systems prescribed should be checked for functionality during mobilization of equipment on board the survey vessel. The functionality checks should include but not be limited to the signal response of sensors, the data-acquisition system, and a wet test of essential subsea equipment.

In-situ equipment with electronic transmission should be designed to sustain the water pressures expected in the field.

During testing, zero readings of all sensors should be recorded before and after each test. The specifications for in-situ test equipment are made for the most commonly used tests. For other in-situ testing, equipment specifications and procedures should be established prior to mobilization.

Records of experience with the use of the equipment, routines, and procedures for interpretation of measurements for assessment of soil parameters should be documented and should be made available on request

In most cases, the in-situ test tool can be inserted into the soils from the sea bed to either a preset depth or to refusal due to limitation in pushing force, capacity of load sensor(s), or other factors, which is called *seabed mode*, or from the bottom of the borehole to either a preset depth or to refusal due to limitation in pushing force, capacity of load sensor(s), or other factors, which is called *drilling mode*.

The drilling of the borehole should be carried out in such a way that the disturbance to the soil below the drill bit is minimized. To avoid any disturbed zone below the drill bit, the in-situ test tool should penetrate at least 1 m if soil strength and density allow. The disturbed zone can be assessed by continuous CPT/CPTU (see next section) penetrated to approximately 3 m below the drill bit.

6.4.1 Cone penetration test

The cone penetration test (CPT; Dutch cone) is modified to piezocone penetration depth. The piezocone test (CPT testing that also gathers piezometer data, called *CPTU testing*) is a CPT with additional measurement of the pore water pressure at one or more locations (U_1 , U_2 , and U_3) on the penetrometer surface, as shown in [Figure 6.1](#).

The CPT is an in-situ testing method for determining the geotechnical engineering properties of soils and for delineating soil stratigraphy. It was initially developed in the 1950s at the Dutch Laboratory for Soil Mechanics in Delft to investigate soft

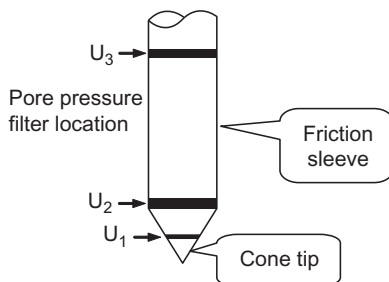


Figure 6.1 Cone penetration testing.

soils (which is why it has also been called the *Dutch cone test*). Today, the CPT is one of the most used and accepted in-situ test methods for soil investigation worldwide.

The test method consists of pushing an instrumented cone, with the tip facing down, into the ground at a controlled rate, usually 20 mm/s. The resolution of the CPT in delineating stratigraphic layers is related to the size of the cone tip, with typical cone tips having a cross-sectional area of either 1000 or 1500 mm², corresponding to diameters of 36 and 44 mm.

The American Society for Testing and Materials (ASTM) presents the apparatus and test procedure for the CPT and the measurement of q_c . In particular, [ISO \(2005\)](#) prescribes cones with a base area in the range of 500 mm² to 2000 mm² and a penetration rate of 20 ± 5 mm/s.

It is noted that the CPT-based design methods were established for cone resistance values up to 100 MPa. Caution should be used when applying the methods to sands with higher resistances.

The early applications of CPT mainly determined soil bearing capacity. The original cone penetrometers involved simple mechanical measurements of the total penetration resistance to pushing a tool with a conical tip into the soil. Different methods were employed to separate the total measured resistance into components generated by the conical tip due to the tip friction and friction generated by the rod string.

In the 1960s, a friction sleeve was added to quantify the friction component and aid in determining soil cohesive strength ([Begemann, 1965](#)). Electronic measurements began in 1948 and improved further in the early 1970s ([de Reister, 1971](#)).

Most modern electronic CPT cones now also employ a pressure transducer with a filter to gather pore water pressure data. The filter is usually located either on the cone tip (the so-called U_1 position), immediately behind the cone tip in the most common U_2 position, or behind the friction sleeve, which is the U_3 position. Pore water pressure data aid in stratigraphy and is primarily used to correct tip friction values. CPT and CPTU testing equipment generally advance the cone using hydraulic rams mounted on a heavily ballasted vehicle or using screwed-in anchors as a counterforce.

CPT for geotechnical applications was standardized in 1986 by ASTM Standard D-3441 ([ASTM, 2004](#)). ISSMGE provides international standards for CPT and

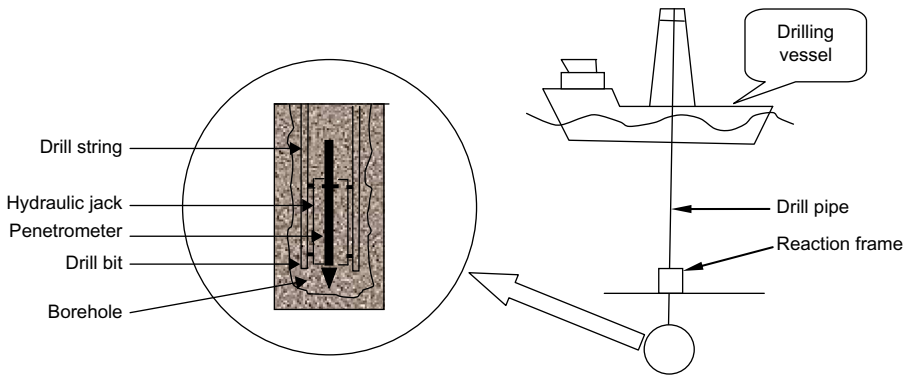


Figure 6.2 Arrangement for cone penetrometer tests.

CPTU. Later, ASTM standards addressed the use of CPT for environmental site characterization and groundwater monitoring activities.

The ability to advance additional in-situ testing tools using the CPT direct-push drilling rig, including the seismic tools, is accelerating this process.

The arrangement for the CPT on the drilling vessel is shown in [Figure 6.2](#).

6.4.1.1 Equipment requirements

The geometry of the cone penetrometer, with tip, sleeve, and pore pressure filters, should follow [IRTP \(1999\)](#), from which the following is extracted: (1) The penetrometer tip and adjoining rods should have the same diameter for at least 400 mm behind the tip, and (2) the cone should have a nominal cross-section area, A_c , of 1000 mm^2 , with $35.3 \text{ mm} \leq d_c \leq 36.0 \text{ mm}$ and $24.0 \text{ mm} \leq h_c \leq 31.2 \text{ mm}$, where d_c is the cone diameter and h_c is the height of the conical part.

According to the [IRTP \(1999\)](#), cone penetrometers with a diameter between 25 mm ($A_c = 500 \text{ mm}^2$) and 50 mm ($A_c = 2000 \text{ mm}^2$) are permitted for special purposes, without the application of correction factors. The recommended geometry and tolerances given for the 1000-mm^2 cone penetrometer should be adjusted proportionally to the diameter.

The use of accuracy classes, as required in [IRTP \(1999\)](#), should be adopted. Equipment and procedures to be used should be selected according to the required accuracy class given in [Table 6.1](#). These precautions are critical, to be considered and monitored by the project quality team.

Class 1 is meant for situations where the results will be used for precise evaluation of stratification and soil type as well as parameter interpretation in profiles including soft or loose soils. Class 2 may be considered more appropriate for stiff clays and sands. For Class 3, the results should be used only for stratification and parameter evaluation in stiff or dense soils.

Table 6.1 Accuracy classes

Class	Measured parameter	Minimum allowable accuracy	Maximum length between measurements, mm
1	Cone resistance	50 kPa or 3%	20
	Sleeve friction	10 kPa or 10%	
	Pore pressure	5 kPa or 2%	
	Inclination	2 degrees	
	Penetration	0.1 m or 1%	
2	Cone resistance	200 kPa or 3%	20
	Sleeve friction	25 kPa or 15%	
	Pore pressure	25 kPa or 3%	
	Inclination	2 degrees	
	Penetration	0.2 m or 2%	
3	Cone resistance	400 kPa or 5%	50
	Sleeve friction	50 kPa or 15%	
	Pore pressure	50 kPa or 5%	
	Inclination	5 degrees	
	Penetration	0.2 m or 2%	

6.4.1.2 CPT results

The reporting of results from CPTs should comply with [IRTP \(1999\)](#).

For each cone test, the following information should be reported offshore after each test and in the field report (either in tabular form or in the CPT profiles):

- Location.
- Test number.
- Coordinates of test location.
- Date of test.
- Cone serial number.
- Cone geometry and dimensions, with position and dimensions of filter stone.
- Capacity of sensors (tip, pore pressure, and sleeve friction).
- Calibration factors used.
- Zero readings of all sensors before and after each test, either at sea floor or bottom of borehole.
- Observed wear or damage on tip or sleeve.
- Penetration rate.
- Any irregularities during testing.
- Theoretical effective (net) area ratio of tip and friction sleeve.
- water depth to sea floor during test.
- corrections due to tidal variations, if any.
- observed sinking in of the frame.
- the inclination of the cone penetrometer to vertical axis, for a maximum penetration depth spacing of 1 m (for sea-bed testing only).

The measured results (in engineering units) should be presented in digital form, as:

- Depth of penetration (in m).
- Cone tip resistance (in MPa).
- Pore pressure(s) (in MPa).
- Sleeve friction (in kPa).
- Total thrust during test (in kN).

A graphical representation of the results from CPTs in the field (offshore) should be presented. If not otherwise agreed on, the depth scale should be 1 m (field) = 10 mm (plot).

The zero reference for seabed CPTs should be the sea bottom, and for down-hole CPTs, the bottom of the borehole. The selection of the scale for presenting the measured cone resistance, pore pressure, and friction should be reasonable to suit the soil conditions.

The output data are presented in a graph, as shown in [Figure 6.3](#).

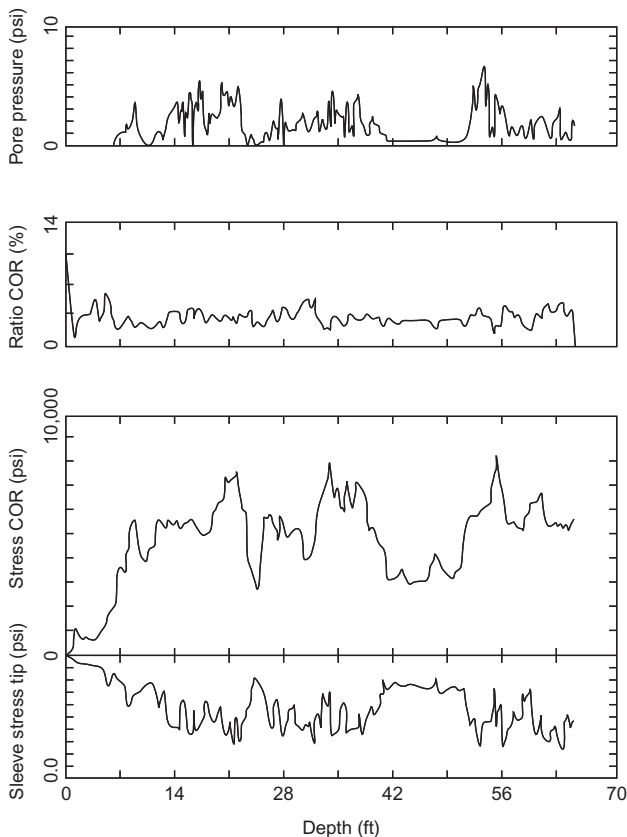


Figure 6.3 Sample of CPT data results.

6.4.2 Field vane test

Vane blades should be rectangular, as defined in ASTM D-2573-01 or BS 5930:1999.

Shear strengths are given in Table 6.2, based on the geometrical dimensions of the vane blades. The vane test in the borehole offshore is presented in Figure 6.4.

As a normal procedure before a vane test is started, the vane blade should be pushed at least 0.5 m below the bottom of the borehole. The pushing rate should be less than 25 mm/s. The time from the instant when the desired test depth has been reached to the beginning of the test (waiting time) should be 2–5 min.

The rotation of the vane should be smooth, and for the initial test (undisturbed), it should be 6–12° per min.

To measure remolded shear strength, the vane should be rotated at least 10 times at a rate ≥ 4 rpm (rev/min) and until a constant torque over 45° of continuous rotation has been reached. At the end of the rapid rotations, the remolded shear strength should be measured without delay, with a rotation rate equal to that used for intact shear strength.

Table 6.2 Measuring range with different vane blade dimensions

Measuring range of s_u	Vane height, mm	Vane width, mm	Vane blade thickness, mm
0–50	130	65	2
30–100	110	55	2
80–250	80	40	2

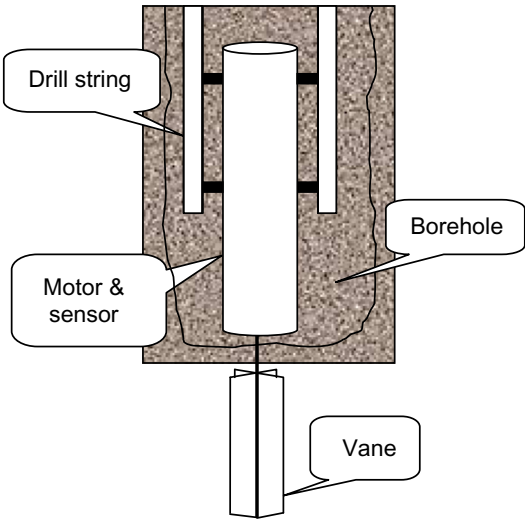


Figure 6.4 Vane test.

It is also possible to do vane tests in the seabed mode. The depth intervals between tests should be at least 0.5 m.

The insertion method and test procedure to be used should be described, giving particular information about the method for insertion and penetration of vanes below the bottom of the borehole, possible rotation rates available, and the method for providing torque and reaction.

The data-acquisition system should be such that the overall accuracy is maintained. Taking into account all sources of error, including the data-acquisition system, the uncertainty in the measured torque should not exceed the smaller of 5% of measured value or 2% of the maximum value of the measured torque of the layer under consideration.

The resolution of the measured result should be within 2% of the measured value.

During testing, the data-acquisition system should allow for real-time inspection of measured results in both digital and graphic form.

It is very important that, at least every year and before each project, the sensor for measuring the torque during vane testing be calibrated. If the sensor is loaded close to its maximum or any damage is suspected, it should be checked and recalibrated. In addition, function checks should be carried out in the field.

For each vane test, the following information should be given:

- Site area.
- Date of test.
- Operator.
- Boring number/test identification.
- Water depth at test location.
- Dimensions of vane.
- Depth below sea bottom to vane tip.
- Depth below bottom of borehole to vane tip.
- Rate of rotation.
- The complete curve of torque versus rotation (degrees).
- Time to failure.
- Any irregularities during testing.
- Formula used to calculate the vane undrained shear strength, s_{uv} , including the assumption made for shear stress distribution on ends of the vane.

6.5 Soil properties

In soil mechanics, the equilibrium and movement of soil bodies is studied, where soil is understood to be the weathered natural material in the upper layers (the upper 20 to 100 m) of the earth's crust. This material may be gravel, sand, clay, peat, or some other soft and loose granular medium. The characteristics of these materials are quite different from those of artificial (human-made) materials, such as steel or concrete. Human-made materials usually are much more consistent than soils and exhibit relatively simple, linear, mechanical behavior, at least if the deformations are not too large.

The mechanical properties of soils are usually strongly nonlinear, with the material exhibiting irreversible plastic deformations when loaded and unloaded, even at low stress levels, and often showing anisotropic behavior, creep, and such typical effects as a volume change during shear. This mechanical behavior of soil is also difficult to predict, because the structure of the soil may be highly inhomogeneous, as a result of its geological history; and it is often not possible to determine the detailed behavior of the soil by tests in the laboratory or in situ.

The behavior of soils may be further complicated by the presence of water in the pores. This relatively stiff fluid in the pores may prevent or retard volume deformations.

For all these reasons, the characterization of the mechanical behavior of soils is often done in a schematic way only, and its form is adapted to the particular type of problem under consideration.

The soil layers have a dominating effect, whereas the settlement of an embankment is mainly governed by the deformation properties of the soil, including creep.

Therefore, in soil mechanics, the range of applicability of a certain parameter is often restricted to a limited class of problems. Many properties cannot be used outside their intended field of application. Nevertheless, various properties may all derive from such common basic phenomena as interparticle friction or the structure of a granular medium, so that good correlations between certain properties may well exist. In this chapter, some of these properties are reviewed and some correlations are discussed. It should be noted that, in engineering practice, nothing can beat the results of experimental determination of the soil parameters in situ or in the laboratory. A correlation may at best give a first estimate of the order of magnitude of a parameter.

6.5.1 Strength

Soils usually cannot transfer stresses beyond a certain limit. This is called the *strength* of the soil. The shear strength of soils is usually expressed by Coulomb's relation between the maximum shear stress and the effective normal stress:

$$\tau_{\max} = c + \sigma' \tan \phi \quad (6.1)$$

where c is the cohesion and ϕ is the friction angle. For sand, c is usually negligible, so that ϕ is the only strength parameter.

For clay, it is often most relevant to consider the strength in undrained conditions, during which the effective stress remains constant. The undrained shear strength is usually denoted by s_u , and it is often considered irrelevant to what degree this is to be attributed to cohesion or friction. The shear strength parameters can be determined in the laboratory, for instance, by triaxial testing.

A simple and useful in-situ test is the cone penetration test, in which a cone is pushed into the ground using hydraulic pressure equipment, while recording the stress at the tip of the cone and often also the friction along the lower part of

the shaft. The test is used in the Netherlands as a model test for a pile foundation, and the results are used directly to determine the bearing capacity of end-bearing piles, using simple scale rules.

The CPT can also be used to estimate the strength of a soil, however, by using certain correlations. For a penetration test in sand, for instance, the bearing capacity of the cone, q_c , is, according to Brinch Hansen's formula:

$$q_c = s_q N_q \sigma'_\nu = s_q N_q \gamma' z \quad (6.2)$$

where s_q is a shape factor to express effective weight of the overburden, for which one may use the formula:

$$s_q = 1 + \sin \phi \quad (6.3)$$

The z is the depth, and N_q is a dimensionless constant for which theoretical analysis has given the value as in the following equation:

$$N_q = \frac{1 + \sin \phi}{1 - \sin \phi} \exp(\pi \tan \phi) \quad (6.4)$$

The predicted cone resistances q_c for various types of sand at a depth of 10 m and 20 m are shown in Table 6.3, assuming that $\gamma = 10 \text{ kN/m}^3$. The values in the table are indeed often observed for sand layers at these depths. They may also be used inversely: if a certain cone resistance is observed, it is indicative for a certain type of material.

For a CPT in clay soil, the Brinch Hansen formula can be used to correlate the CPT result to the undrained shear strength.

The general Brinch Hansen formula is

$$q_c = s_c N_c c + s_q N_q \sigma'_\nu \quad (6.5)$$

Because the test is performed very quickly, the soil behavior can be considered to be undrained.. The values for the coefficients can then be taken as $N_c = 5.14$, $N_q = 1$, $s_c = 1.3$, $s_q = 1$. Equation (6.5) now reduces to

$$q_c = 6.7 s_u + \sigma'_\nu \quad (6.6)$$

where the cohesion c has been interpreted as the undrained shear strength s_u .

Table 6.3 Guidance for cone resistance in sand

Soil type	ϕ	N_q	$q_c, z = 10 \text{ m, MPa}$	$q_c, z = 20 \text{ m; e, MPa}$
Loose sand	30°	18.4	2.8	5.5
Medium dense sand	35°	33.3	5.2	10.5
Very dense sand	40°	64.2	10.5	21.1

The undrained shear strength of normally consolidated clays depends on the vertical stress σ'_ν . A relationship that is often used is the correlation proposed by Ladd,

$$s_u = 0.22\sigma'_\nu \tag{6.7}$$

Substitution of this result into [equation \(6.6\)](#) gives

$$q_c \approx 11s_u \tag{6.8}$$

For a soft clay, with $s_u = 20$ kPa, the order of magnitude of the cone resistance would be 220 kPa $\approx$ 0.2 MPa. Such values are indeed often observed. Again, they may also be used to estimate the undrained shear strength from CPT data.

[Table 6.4](#) provides guidance to cohesive soil characteristics based on the rule of thumb.

The relative density for cohesionless soil can be predicted using [Table 6.5](#) as a guideline.

The approximate soil parameters for different types of soil are illustrated in [Table 6.6](#).

Table 6.4 Consistency of cohesive soil

Consistency	Unconfined compressive strength, tons/ft ²	Rule-of-thumb test
Very soft	0–0.25	Core (height twice diameter) sags under own weight
Soft	0.25–0.5	Can be pinched in two by pressing between thumb and finger
Firm	0.5–1.0	Can be imprinted easily with finger
Stiff	1.0–2.0	Can be imprinted with considerable pressure from fingers
Very stiff	2.0–4.0	Barely can be imprinted by pressure from fingers
Hard	>4.0	Cannot be imprinted by fingers

Table 6.5 Degree of compactness for cohesionless soil

Degree of compactness	Relative density, %
Very loose	0–15
Loose	15–35
Medium dense	35–65
Dense	65–85
Very dense	85–100

Table 6.6 Approximate property values for different soil types

Soil type	Undrained shear strength, S_u , in kPa	Effective cohesion, c' , in kPa	Friction angle, f' , in degrees	Saturated density, D_s , in t/m^3	Voids ratio
Soft to firm clay	10–50	5–10	19–24	1.4–1.8	
Stiff clay	50–100	10–20	22–29	1.8–1.9	
Very stiff to hard clay	100–400	20–50	27–31	1.9–2.2	
Silt	10–50	—	27–35	1.7–2.3	1.1–0.3
Loose sand	—	—	29–30	1.7–1.8	1.1–0.8
Medium dense sand	—	—	30–40	1.8–2.0	0.8–0.5
Dense sand	—	—	35–45	2.0–2.3	0.5–0.2
Gravel	—	—	35–55	1.7–2.4	1.1–2.2

6.5.2 Soil characterization

A key part of developing realistic analytical models to evaluate cyclic loading effects on piles is the characterization of soil-pile interaction behavior. High-quality in-situ, laboratory, and model-prototype pile-load tests are essential in such characterizations. In developing soil-pile interaction characterizations, it is important that pile installation and pile-loading conditions be integrated into the testing programs.

According to [McClelland and Ehlers \(1986\)](#), in-situ tests, such as vane shear, cone penetrometer, and pressure meter tests, can provide important insights into in-place soil behavior and stress-strain properties.

Both low- and high-amplitude stress-strain properties can be developed. Long-term static and creep loadings or short-term dynamic, impulsive, and cyclic loadings sometimes can be simulated with in-situ testing equipment.

Laboratory tests on representative soil samples permit a wide variety of stress-strain conditions to be simulated and evaluated. Soil samples can be modified to simulate pile-installation effects, such as remolding and reconsolidating, to estimate in-situ stresses. The samples can be subjected to different boundary conditions, such as triaxial, simple shear, and interface shear, and to different levels of sustained and cyclic shear time histories to simulate in-place loading conditions.

Another important source of data to develop soil characterizations for cyclic loading analyses are tests on model and prototype piles. Based on [Bogard et al. \(1985\)](#)

and [Karlsrud and Haugen \(1985\)](#), model piles can be highly instrumented, and repeated tests can be performed in soils and for a variety of loadings.

Geometric scale, time scale, and other modeling effects should be carefully considered in applying results from model tests to prototype behavior analyses. As discussed by [Pelletier and Doyle \(1982\)](#) and [Arup et al. \(1986\)](#), the data from load tests on prototype piles are useful for calibrating analytical models.

Such tests, even if not highly instrumented, can provide data to guide development of analytical models. These tests can also provide data to verify results of soil characterizations and analytical models.

Prototype pile-load testing, coupled with in-situ and laboratory soil testing and realistic analytical models, can provide an essential framework for making realistic evaluations of the responses of piles to cyclic axial loadings.

The foundation should be designed to carry static, cyclic, and transient loads without excessive deformation or vibrations in the platform. Special attention should be given to the effects of cyclic and transient loading on the strength of the supporting soils as well as on the structural response of piles.

It is very important to consider the possibility of movement of the sea floor against the foundation members, and the forces caused by such movements, if anticipated, should be considered in the design.

6.6 Pile foundations

Offshore structure platforms commonly use open driven piles. These piles are usually driven into the seabed with impact hammers, which use steam, diesel fuel, or hydraulic power as the source of energy. Therefore, the pile wall thickness should be adequate to resist axial and lateral loads as well as the stresses during pile driving.

According to [Smith \(1962\)](#), it is possible to predict approximately the stresses during pile driving using the principles of one-dimensional elastic stress wave transmission by carefully selecting the parameters that govern the behavior of soil, pile, cushions, cap block, and hammer.

This approach may also be used to optimize the pile hammer cushion and cap block using the computer analysis, commonly known as *wave equation analysis*. The design penetration of driven piles should be determined, rather than correlation of pile capacity with the number of blows required to drive the pile a certain distance into the seabed.

If a pile stops before it reaches design penetration, one or more of the following actions can be taken:

1. Review of all aspects of hammer performance, possibly with the aid of hammer and pile-head instrumentation, which may identify problems that can be solved by improving hammer operation and maintenance or by the use of a more powerful hammer.
2. Reevaluation of design penetration by reconsideration of loads, deformation, and required capacities of both individual piles and the foundation as a whole, which may identify

reserve capacity available. An interpretation of driving records in conjunction with the instrumentation mentioned previously may allow design soil parameters or stratification to be revised and pile capacity to be increased.

3. Modifications to piling procedures (usually, the last course of action), which may include one of the following: The soil plug inside the pile is removed by jetting and air lifting or by drilling to reduce pile-driving resistance. If plug removal results in inadequate pile capacities, the removed soil plug should be replaced by a gravel grout or concrete plug having sufficient load-carrying capacity to replace that of the removed soil plug. Attention should be paid to plug and pile load-transfer characteristics. Note that plug removal may not be effective in some circumstances, particularly in cohesive soils.

Soil below the pile tip is removed by drilling an undersized hole or by lowering jetting equipment through the pile, which acts as the casing pipe for the operation. The effect on pile capacity of drilling an undersized hole is unpredictable unless there has been previous experience under similar conditions.

According to API RP2A (2007), jetting below the pile tip should be avoided because of the unpredictability of the results.

A first stage or outer pile is driven to a predetermined depth, the soil plug is removed, and a second stage or inner pile is driven inside the first-stage pile. In this case, the grouting is inserted in the annulus between the two piles to provide load transfer between the two piles and to develop composite action.

The piles of the offshore structures are exposed to static and cyclic loading, and the loads are axial and lateral. So, all these load effects should be considered in the pile design.

6.6.1 Pile capacity for axial loads

On the basis of API RP2A (2007), the ultimate static axial capacity (Q) of an open-ended pipe pile in compression is given by the equation

$$Q_t = Q_f + Q_s + \{\text{small values from } Q_{fi} \text{ or } Q_{sp}\} \quad (6.9)$$

$$Q_t = f \cdot A_s + A_a \cdot Q + A_p \cdot q \quad (6.10)$$

where A_p is the area of pile or

$$Q_t = f \cdot A_s + A_a \cdot Q + f A_{si} \quad (6.11)$$

where

- Q_t = the total pile resistance
- Q_f = total outside shaft resistance
- Q_s = end-bearing capacity of the annulus
- Q_{fi} = total inside shaft resistance
- Q_{sp} = bearing capacity of the soil beneath the plug
- f = unit skin friction capacity (in kPa)
- A_s = outside surface area of pile (in m^2)

q = unit end-bearing capacity (in kPa)

A_a = the area of the pile annulus (in m^2)

A_{si} = inside surface area of pile (in m^2)

In computing pile loading and capacity, the weight of the pile-soil plug system and hydrostatic uplift should be considered.

In determining the load capacity of a pile, consideration should be given to the relative deformation between the soil and the pile, as well as the compressibility of the soil-pile system.

Coyle and Reese (1966), Murff (1980), and Randolph (1983) discuss skin friction and assume that the maximum skin friction along the pile and the maximum end bearing are mobilized simultaneously. However, the ultimate skin friction increments along the pile are not necessarily directly additive, nor is the ultimate end bearing necessarily additive with the ultimate skin friction. In some circumstances, this effect may result in the capacity being less than that given by equation (6.11).

In such cases, a more explicit consideration of axial pile performance effects on pile capacity may be warranted. Pile sizing should be based on what experience has shown can be installed consistently, practically, and economically under similar conditions with the installation equipment being used. Alternatives for possible remedial action in the event design objectives cannot be obtained during installation should also be investigated and defined prior to construction.

For the pile system, the pile-capacity factor of safety is defined in Table 6.7, according to API RP2A (2007). The allowable skin friction values on the pile section on the upper surface on the pile should be discounted in computing skin friction resistance, Q_f . The end-bearing area of a pilot hole, if drilled, should be discounted in computing total bearing area.

Table 6.7 Design parameter guide for cohesionless siliceous soil (based on API RP2A)

Soil description	Soil condition	Shaft friction factor	Limited shaft friction values, f , kPa	End-bearing factor, N_q	Limited unit end-bearing values, q , MPa
Sand	Very loose				
Sand	Loose				
Sand-silt	Loose				
Silt	Medium dense				
Silt	Dense				
Sand-silt	Medium dense	0.29	67	12	3
Sand	Medium dense	0.37	81	20	5
Sand-silt	Dense	0.37	81	20	5
Sand	Dense	0.46	96	40	10
Sand-silt	Very dense	0.46	96	40	10
Sand	Very dense	0.56	115	50	12

6.6.1.1 Skin friction and end bearing in cohesive soils

For pipe piles in cohesive soils, the shaft friction, f (in kPa), at any point along the pile may be calculated by

$$f = \alpha c \quad (6.12)$$

where α = a dimensionless factor and c = undrained shear strength of the soil at the point in question.

The factor α can be computed by

$$\begin{aligned} \alpha &= 0.5\psi - 0.5\psi \leq 1.0 \\ \alpha &= 0.5\psi - 0.25\psi > 1.0 \end{aligned} \quad (6.13)$$

with the constraint that $\alpha \leq 1.0$, where $\psi = c/p'$ for the point in question and p' is the effective overburden pressure at the point in question (in kPa). For underconsolidated clays, clays with excess pore pressures undergoing active consolidation, α can usually be taken as 1.0.

A discussion of appropriate methods for determining the undrained shear strength, c , and effective overburden pressure, p' , including the effects of various sampling and testing procedures, is important. Due to the lack of pile-load tests in soils having c/p' ratios greater than 3, [equation \(6.13\)](#) should be applied with some engineering judgment for high c/p' values. Similar judgment should be applied for deep-penetrating piles in soils with high undrained shear strength c , where the computed shaft frictions, f , using [equation \(6.14\)](#), are generally higher than previously specified in API RP2A. For very long piles, some reduction in pile capacity would happen, because the shaft friction may reduce to some lesser residual value on continued displacement.

For piles end bearing in cohesive soils, the unit end bearing, q (in kPa), may be computed by

$$q = 9c \quad (6.14)$$

It is obvious that, in open-driven piles, the shaft friction, f , acts on both the inside and outside of the pile. The total resistance is the sum of the external shaft friction, the end bearing on the pile wall annulus, and the total internal shaft friction or the end bearing of the plug, whichever is less.

For piles considered to be plugged, the bearing pressure may be assumed to act over the entire cross section of the pile. For unplugged piles, the bearing pressure acts on only the pile wall annulus. Whether a pile is considered plugged or unplugged may be based on static calculations. For example, a pile could be driven in an unplugged condition but act plugged under static loading.

For piles driven in undersized drilled holes, piles jetted in place, or piles drilled and grouted in place, the selection of shaft friction values should take into account the soil disturbance resulting from installation. In general, f should not exceed

values for driven piles; however, in some cases, for drilled and grouted piles in over consolidated clay, f may exceed these values.

In determining f for drilled and grouted piles, the strength of the soil-grout interface, including potential effects of drilling mud, should be considered. A further check should be made of the allowable bond stress between the pile steel and the grout, which was discussed by Kraft and Lyons (1974).

In layered soils, shaft friction values, f , in the cohesive layers should be as given in equation (6.12). End-bearing values for piles tipped in cohesive layers with adjacent weaker layers may be as given in equation (6.14), assuming that the pile achieves penetration of two to three diameters or more into the layer in question and the tip is approximately three diameters above the bottom of the layer, to preclude punch-through.

Where these distances are not achieved, some modification in the end-bearing resistance may be necessary. Where adjacent layers are of comparable strength to the layer of interest, the proximity of the pile tip to the interface is not a concern.

Example 6.1

Calculate the capacity for plug pile for the following data.

Pile diameter = 48 in. = 1.219
 Pile thickness = 25.4 mm
 Pile length = 40 m
 Skin friction $f = 70$ kPa
 End bearing, $Q = 2.64$ MPa at the last layer

Solution

Base area = 1.167 m^2
 Surface area of the pile, $A_s = 3.83 \times 40 = 153.18 \text{ m}^2$
 Total pile capacity = $153.18 \times 70,000 + 1.167 \times 2,640,000 = 11,002,680 \text{ N} = 1.1 \text{ MN}$

Example 6.2

Calculate the capacity for the pile based on API for the following data.

Pile diameter = 60 in. = 1.524
 Pile thickness = 22 mm
 Pile length = 40 m
 Skin friction, $f = 70$ kPa
 End bearing, $Q = 2.64$ MPa at the last layer

Solution

Base area = 1.824 m^2
 Area of the annulus = 0.104 m^2
 Surface area of the pile, $A_s = 4.788 \times 40 = 191.51 \text{ m}^2$
 Pile internal diameter = 1.48 m
 Surface area of pile internal = $4.65 \times 40 = 185.98 \text{ m}^2$

In case of a plugged pile,

Total pile capacity = $(191.51 \times 70,000 + 1.824 \times 2,640,000) \times 10^{-6} = 18.2 \text{ MN}$

In case of an unplugged pile,

Total pile capacity = $[(191.51 + 185.98) \times 70,000 + 0.104 \times 2,640,000] \times 10^{-6} = 26.7 \text{ MN}$

The design pile capacity = 18.2 MN

Example 6.3

Calculate pile capacity at different soil strata using the following input data:

OD = 1.3 m

The pile depth = 33 m

Total base area = 1.327 m^2

Outside shaft area/ $m' = 4.08 \text{ m}^2$

Cohesive soil. $k_o = 1$

Noncohesive soil. $k_o = 0.5$

The soil consists of nine layers, nc for noncohesive soil, and c for cohesive soil.

Solution Table 6.8 presents the solution. You can get the Excel sheet by contacting the website www.elreedy.comli.com.

6.6.1.2 Shaft friction and end bearing in cohesionless soils

This section provides a simple method for assessing pile capacity in cohesionless soils. There are reliable methods for predicting pile capacity that are based on direct correlations of pile unit friction and end-bearing data with cone penetration test results. The CPT-based methods are considered fundamentally better, have shown statistically closer predictions of pile-load test results, and although not required, they are, in principle, the preferred methods. CPT-based methods also cover a wider range of cohesionless soils. However, offshore experience with CPT methods is limited, and hence, more experience is needed before they are recommended for routine design. CPT-based methods should be applied only by qualified engineers who are experienced in the interpretation of CPT data and understand the limitations and reliability of the methods. Following installation, pile-driving instrumentation data may be used to give more confidence in predicted capacities.

For pipe piles in cohesionless soils, the unit shaft friction at a given depth, f , may be calculated by

$$f = \beta p' \quad (6.15)$$

where β = a dimensionless shaft friction factor and p' = effective overburden pressure at the depth in question.

Table 6.7 may be used for selection of β values for open-ended pipe piles driven unplugged if other data are not available. Values of β for full-displacement piles (i.e., driven fully plugged or closed ended) may be assumed to be 25% higher than those given in Table 6.11. For long piles, f may not increase linearly with the overburden pressure, as implied by equation (6.15). In such cases, it may be appropriate to limit f to the values given in Table 6.9.

Table 6.8 Pile capacity calculation

Layer number, enter c or nc	Depth, m	Soil p for s'_v , kN/m ³	s'_v , kPa	p'_o , kPa	Cohesive soil			Non cohesive soil				Q shaft for layer, MN*	Q base bottom, MN**	Q total,
					C_u , kPa	Skin friction, f , kPa	End bearing, q , MPa	Soil-pile friction, d , °	N_q	Skin friction, f , kPa	End bearing, q , MPa			
1 c	0	9.2	0	0	115	0.0	1.035	0	0	NA	NA	0.47	1.37	1.84
	5	9.2	46	46	115	45.7	1.035	0	0	NA	NA			
2 c	5	9.2	46	46	112.5	45.0	1.0125	0	0	NA	NA	0.79	1.34	2.60
	9	9.2	83	83	112.5	52.1	1.0125	0	0	NA	NA			
3 c	9	9.2	83	83	100	47.7	0.9	0	0	NA	NA	1.07	1.19	3.52
	14	9.2	129	129	100	56.7	0.9	0	0	NA	NA			
4 c	14	9.2	129	129	125	63.4	1.125	0	0	NA	NA	2.18	1.49	6.00
	21.5	9.2	198	198	125	78.6	1.125	0	0	NA	NA			
5 nc	21.5	9.5	198	132	0	NA	NA	30	20	60.9	2.64	0.52	3.84	8.86
	23.5	9.5	217	145	0	NA	NA	30	20	66.8	2.89			
6 nc	23.5	9.5	217	145	0	NA	NA	31	20	69.5	2.89	0.44	4.09	9.55
	25	9.5	231	154	0	NA	NA	31	20	74.0	3.08			
7 c	25	9.5	231	231	140	89.9	1.26	0	0	NA	NA	0.75	1.67	7.88
	27	9.5	250	250	140	93.6	1.26	0	0	NA	NA			
8 nc	27	9.5	250	167	0	NA	NA	32	20	83.3	3.33	0.53	4.68	11.41
	28.5	9.5	264	176	0	NA	NA	32	20	88.1	3.52			
9 c	28.5	9.5	264	264	85	74.9	0.765	0	0	na	na	1.43	1.02	9.18
	33	9.5	307	307	85	80.8	0.765	0	0	na	na			

*Skin friction by selecting its value from cohesive (c) or non cohesive (nc) soil layer multiply by surface area.

**End bearing calculated by selecting its value from cohesive (c) or non cohesive (nc) soil layer multiply by pile area.

Table 6.9 Pile-capacity factor of safety in API RP2A (2007)

Load condition	Factor of safety
Design environmental conditions with appropriate drilling loads	1.5
Operating environmental conditions during drilling operations	2.0
Design environmental conditions with appropriate producing loads	1.5
Operating environmental conditions during producing operations	2.0
Design environmental conditions with minimum loads (for pullout)	1.5

For pile end bearings in cohesionless soils, the unit end bearing q may be calculated by

$$q = Nq p' \quad (6.16)$$

where Nq = a dimensionless bearing capacity factor and p' = effective overburden pressure at the depth in question. Recommended Nq values are also presented in Table 6.7.

For long piles, q may not increase linearly with the overburden pressure, as implied by equation (6.16). In such cases, it may be appropriate to limit q to the values given in Table 6.9. For plugged piles, the unit end bearing q acts over the entire cross section of the pile. For unplugged piles, q acts on the pile annulus only. In this case, additional resistance is offered by friction between the soil plug and the inner pile wall.

Whether a pile is considered to be plugged or unplugged may be based on static calculations using a unit skin friction on the soil plug equal to the outer skin friction. It is noted that a pile could be driven in an unplugged condition but can act plugged under static loading.

The design parameters in Table 6.9 are just a guide from API RP2A, and detailed information must be obtained from the CPT results, strength tests, and other soil and pile response tests.

Olson (1987) compared the load test data for piles in sand (obtained by measuring the axial load capacities for open steel piles) and the calculated capacity from API RP2A. Studies done by Lehane et al. (2005a, 2005b) indicate that variability in capacity predictions using the API calculation method may exceed those for piles in clay. These researchers also indicated that the calculation method is conservative for short offshore piles (short = piles less than 45 m, or 150 ft, long) in dense to very dense sands loaded in compression and may not be conservative in all other conditions. In unfamiliar situations, the designer may want to account for this uncertainty through a selection of conservative design parameters or by going toward higher factors of safety.

For soils that do not fall within the ranges of soil density and descriptions given in Table 6.7 or for materials with unusually weak grains or compressible structure, Table 6.7 may not be appropriate for selection of design parameters. For example, very loose silts or soils containing large amounts of mica or volcanic grains may require special

laboratory or field tests for selection of design parameters. Of particular importance are sands containing calcium carbonate, which are found extensively in many ocean areas. Experience suggests that driven piles in these soils may have substantially lower design strength parameters rather than those described in Table 6.9.

Drilled and grouted piles in carbonate sand may have significantly higher capacities than driven piles and have been used successfully in many areas with carbonate soils. The characteristics of carbonate sands are highly variable and local experience should dictate the design parameters selected. For example, experience suggests that capacity is improved in carbonate soils of high densities and higher quartz contents.

Cementation may increase end-bearing capacity but result in a loss of lateral pressure and a corresponding decrease in frictional capacity.

For piles driven in undersized drilled or jetted holes in cohesionless soils, the values of f and q should be determined by some reliable method that accounts for the amount of soil disturbance due to installation, but they should not exceed values for driven piles. Except in unusual soil types, such as described previously, the f and q values given in Table 6.7 may be used for drilled and grouted piles, with consideration given to the strength of the soil-grout interface.

In layered soils, unit shaft friction values in cohesionless layers and the end-bearing values for piles tipped in cohesionless layers with adjacent layers of lower strength may also be taken from Table 6.7, provided that the pile achieves penetration of two to three diameters or more into the cohesionless layer, and the tip is at least three diameters above the bottom of the layer, to preclude punch-through. Where these pile tip penetrations are not achieved, some modification in the tabulated values may be necessary. Where adjacent layers are of comparable strength to the layer of interest, the proximity of the pile tip to the layer interface is not a concern.

6.6.2 Foundation size

In most cases, in the FEED engineering phase, the pile configuration is defined based on past experience. During selection of the size of the pile foundation, the following items should be considered: diameter, penetration, wall thickness, type of tip, spacing, number of piles, geometry, location, mudline restraint, material strength, installation method, and other parameters as may be considered appropriate.

A number of different analysis procedures may be utilized to determine the requirements of a foundation. At a minimum, the procedure used should properly simulate the nonlinear response behavior of the soil and ensure load-deflection compatibility between the structure and the pile-soil system.

For deflections and rotations of individual piles, the total foundation system should be checked at all critical locations, which may include pile tops, points of contra flexure, mudline, and so forth. Deflections and rotations should not exceed serviceability limits, which would render the structure inadequate for its intended function.

6.6.2.1 *Pile penetration*

The design pile penetration should be sufficient to develop adequate capacity to resist the maximum computed axial bearing and pullout loads with an appropriate factor of safety.

The ultimate pile capacities can be computed in accordance with previous sections or by other methods that are supported by reliable comprehensive data. API RP2A (2007) defined the minimum factor of safety by dividing the ultimate pile capacity into the actual load, as shown in [Table 6.9](#).

Two safety factors in API RP2A depend on the design considering environmental conditions with a 100-year storm wave effect and the operating environmental conditions, which includes the maximum wave height per year.

The provisions of API RP2A (2007) for sizing the foundation pile are based on an allowable stress (working stress) method, except for pile penetration. In this method, the foundation piles should conform to the requirements of specification and design. Any alternative method supported by sound engineering methods and empirical evidence may also be utilized. Such alternative methods include the limit state design approach or ultimate strength design of the total foundation system.

6.6.3 *Axial pile performance*

6.6.3.1 *Static load-deflection behavior*

The static pile axial deflection should be compatible with the structural forces and deflection, so it should be within the service limit. An analytical method for determining axial pile performance is provided in [Meyer, Holmquist, and Matlock \(1975\)](#).

This method uses axial pile shear transition versus local pile deflection (t - z) curves to model the axial support provided by the soil along the sides of the pile. An additional (Q - z) curve is used to model the tip and bearing versus the deflection response. (Methods for constructing t - z and Q - z curves are given later.) Pile response is affected by load directions, load types, load rates, loading sequence, installation technique, soil type, axial pile stiffness, and other parameters. Some of these effects for cohesive soils have been observed in both laboratory and field tests.

In some circumstances, as for soils that exhibit strain-softening behavior and where the piles are axially flexible, the actual capacity of the pile may be less than that given by [equation \(6.9\)](#). In these cases, an explicit consideration of the effects on ultimate axial capacity may be warranted. Note that other factors, such as increased axial capacity under loading rates associated with storm waves, may counteract these effects, as discussed by [Dunnavant et al. \(1990\)](#).

6.6.3.2 *Cyclic response*

Unusual pile loading conditions or limitations on design pile penetrations may warrant detailed consideration of cyclic loading effects.

Cyclic loading, which includes inertial loadings developed by environmental conditions such as storm waves and earthquakes, can have two potentially counter-active effects on the static axial capacity. Repetitive loadings can cause a temporary or permanent decrease in load-carrying resistance or an accumulation of deformation. Rapidly applied loadings can cause an increase in load-carrying resistance or stiffness of the pile. Very slowly applied loadings can cause a decrease in load-carrying resistance or stiffness of the pile. The resultant influence of cyclic loadings are a function of the combined effects of the magnitudes, cycles, and rates of applied pile loads; the structural characteristics of the pile; the types of soils; and the factors of safety used in design of the piles. The design pile penetration should be sufficient to develop an effective pile capacity to resist the design static and cyclic loadings.

The design pile penetration can be confirmed by performing pile response analyses of the pile-soil system subjected to static and cyclic loadings. The pile-soil resistance-displacement t - z and Q - z characterizations are discussed next.

When any of the preceding effects are explicitly considered in pile-response analysis, the design static and cyclic loadings should be imposed on the pile top and the resistance-displacements of the pile determined. At the completion of the design loadings, the maximum pile resistance and displacement should be determined. Pile deformations should meet structure serviceability requirements.

6.6.3.3 Axial load-deflection (t - z and Q - z) data

The pile foundation should be designed to resist static and cyclic axial loads. The axial resistance of the soil is provided by a combination of axial soil-pile adhesion or load transfer along the sides of the pile and end-bearing resistance at the pile tip. The plotted relationship between mobilized soil-pile shear transfer and local pile deflection at any depth is described using a t - z curve. Similarly, the relationship between mobilized end-bearing resistance and axial tip deflection is described using a Q - z curve.

Axial deformation of piles may be modeled in a way similar to the lateral case, to permit stress transfer to be computed and axial pile stiffness to be assessed. For axial loading, t - z curves are used to represent the resistance along the pile shaft, and Q - z curves are introduced to model end bearing. Characteristic shapes of the curves for sand and clay are shown in [Figures 6.5 and 6.6](#) for clay and sand soil, respectively.

Axial support to a pile is provided by surrounding soil, and axial pile deformation at the end of pile depth may be considered to consist of four components: elastic pile deformation, elastic soil deformation, plastic soil deformation, and plastic soil-pile slip deformation. The purpose of t - z curves is to model the latter three components. Q - z curves model elastic and plastic soil deformation around the pile tip. Elastic pile deformation is not directly related to soil characteristics and is modeled in the beam-column representation of the pile.

In the past, t - z curves were based directly on experimental evidence from [Coyle and Reese \(1966\)](#). This led to an adopted standard that peak shaft resistance was

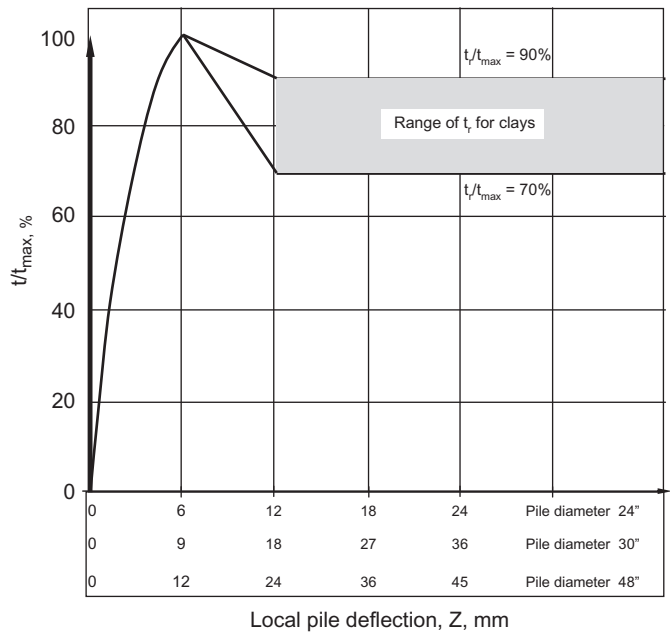


Figure 6.5 The shape of the t-z curve for clay soil for different pile diameters.

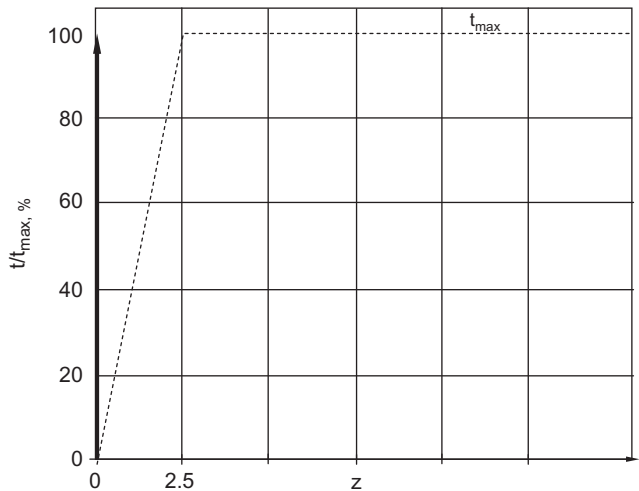


Figure 6.6 The shape of the t-z curve for sand soil for different pile diameters.

mobilized at a vertical relative pile-soil movement (Z_c) of 2.54 mm (0.1 in.) in sand. For clay, the value is 1% of pile diameter. It has been shown theoretically and experimentally that the form of the t - z curve is a function of the pile length and diameter, soil stiffness, and shaft resistance, as discussed by Kraft et al. (1981a, 1981b). However, to account for these characteristics, the average shear modulus of the soil must be known. In sand, the appropriate strain-level shear modulus is known only within an order of magnitude, and hence t - z curves generated by this method will contain considerable uncertainty. However, a parametric study by Meyer et al. (1975) showed that a sixfold variation in soil yield displacement had only a small effect on the predicted pile head displacement.

The data include a peak-residual behavior for t - z curves in clay, the governing parameter of which is the ratio of peak to residual unit skin friction. The recommended range for this parameter is 0.7 to 0.9. Vijayvergiya (1977) indicates that this parameter decreases with an increasing overconsolidation ratio. The peak residual behavior in sand has been adopted as per Wiltsie et al. (1982).

For some projects, in the absence of more definitive criteria, the recommended t - z curves for noncarbonate soils, according to API RP2A, are shown in Figure 6.6.

Table 6.10 presents the relation between the vertical displacement of the pile to the pile diameter and the adhesion between the pile and the soil as a percentage of the total friction capacity. The shape of the t - z curve at displacements greater than $z_{\max}$, as shown in Figure 6.7, should be carefully considered.

The ratio between the residual adhesion to the maximum adhesion between pile and soil, $t_r/t_{\max}$, at the axial pile displacement at which it occurs (z_r) is affected by

Table 6.10 Relation between the ratio of pile deflection to the diameter and the skin friction capacity

Soil type	z/D	Pile axial deflection for different pile diameters, mm			$t/t_{\max}$
		24"	36"	48"	
Clays	0.0016	0.98	1.46	2.0	0.30
	0.0031	1.89	2.83	3.8	0.50
	0.0057	3.5	5.21	7.0	0.75
	0.0080	4.9	7.32	9.8	0.90
	0.0100	6.1	9.14	12.2	1.00
	0.0200	12.2	18.3	24.4	0.70–0.90
	∞				0.70–0.90
Sands	z (mm)				$t/t_{\max}$
	0.000	0	0	0	0.00
	2.5	2.5	2.5	2.5	1.00
	∞				1.00

Note: z = local pile deflection (in mm), D = pile diameter (in mm), t = mobilized soil pile adhesion (in lb/ft² or kPa), and $t_{\max}$ = maximum soil pile adhesion or unit skin friction capacity computed (in lb/ft² or kPa).

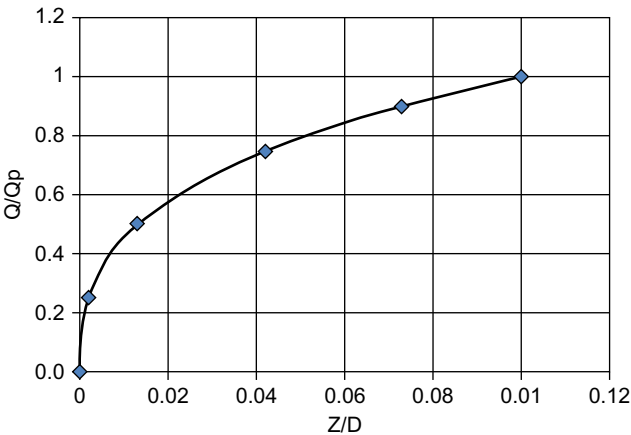


Figure 6.7 Pile tip load displacement (Q - z) curve.

the soil stress-strain behavior, stress history, pipe installation method, pile load sequence, and other factors.

The value of $t_r/t_{\max}$ can range from 70% to 90%. Laboratory, in-situ or model pile tests can provide valuable information for determining values of $t_r/t_{\max}$ and z_r for various soils.

The end-bearing or tip-load capacity should be determined. However, relatively large pile tip movements are required to mobilize the full end-bearing resistance. A pile-tip displacement up to 10% of the pile diameter may be required for full mobilization in both sand and clay soils. In the absence of more definitive criteria, the curve in Figure 6.7 is recommended for both sands and clays.

Table 6.11 presents the relation between the axial displacement of the pile relative to the pile diameter and the end-bearing capacity as a percentage for the total end bearing capacity.

6.6.3.4 Axial pile capacity

A number of studies were performed that were aimed at collecting and comparing axial capacities from relevant pile load tests to those predicted by traditional offshore pile design procedures. Studies like these can be very useful in tempering one’s judgment in the design process. It is clear, for example, that there is considerable scatter in the various plots of measured versus predicted capacities. The designer should be aware of the many limitations of such comparisons when making use of the results. Limitations of particular importance include

1. There is considerable uncertainty in the determination of both predicted capacities and measured capacities. For example, determination of the predicted capacities is very sensitive to the selection of the undrained shear strength profile, which itself is subject to

Table 6.11 Relation between the axial deflection to pile diameter ratio and percentage of end-bearing capacity

z/D	Pile axial deflection for different pile diameters, mm			Q/Q _p
	24"	36"	48"	
0.002	1.2	1.8	2.4	0.25
0.013	7.9	11.9	15.8	0.50
0.042	25.6	38.4	51.2	0.75
0.073	44.5	66.8	89	0.90
0.100	61	91.4	121.9	1.00

Note: z = axial tip deflection (in mm), D = pile diameter (in mm), Q = mobilized end-bearing capacity (in lb or kN) and Q_p = total end bearing (in lb or kN).

considerable uncertainty. The measured capacities are also subject to interpretation as well as possible measurement errors.

2. Conditions under which pile load tests are conducted generally vary significantly from design loads and field conditions. One clear limitation is the limited number of tests on deeply embedded, large-diameter, high-capacity piles. Generally, pile load tests have capacities that are 10% or less of the prototype capacities. [Briaud et al. \(1984\)](#) mentioned that the rate of loading and the cyclic load history are usually not well represented in load tests. According to [Clarke \(1993\)](#), pile load tests are often conducted before full setup occurs, for practical reasons. Furthermore, pile-tip conditions (closed versus open ended) may differ from offshore piles.
3. In most of the studies, an attempt has been made to eliminate the factors thought to be significantly affected by extraneous conditions in load testing, such as protrusions on the exterior of the pile shaft (weld beads, cover plates, etc.), installation effects (jetting, drilled-out plugs, etc.), and artesian conditions, but it is not possible to be absolutely certain in all cases.

The tests most relevant to offshore applications have all been conducted in the United States or Europe. As regional geology and particularly operating experience are considered very important in foundation design, care should be exercised in applying these results to other regions of the world. In addition, the designer should note that certain important tests in silty clays of low plasticity, indicate overprediction of frictional resistance by [equations \(6.12\) and \(6.13\)](#). The reason for the overprediction is not well understood and has been an area of active research. The designer is therefore cautioned that pile design for soils of this type should be given special consideration. Additional considerations that apply to drilled and grouted piles are discussed by [Kraft and Lyons \(1974\)](#) and [O'Neill and Hassan \(1994\)](#).

Pile load tests are commonly used as basis for determining pile load-movement characteristics. As discussed by [El-Reedy \(2011\)](#), in clay, the ultimate capacity of the pile reaches a maximum value at some movement, beyond which there is a gradual drop to a residual value.

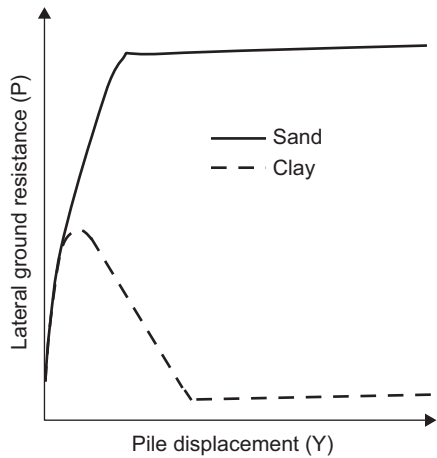


Figure 6.8 Example of typical p - y curve for 36" pile.

The frictional resistance increases rapidly and reaches a maximum value at a very small displacement, referred to as the *critical movement*. However, the point resistance continues to increase beyond this critical movement and tends to reach a maximum value at a relatively larger movement. This maximum value is referred to as the *end-bearing capacity*.

In sand, the ultimate capacity seems to increase and reach a constant value. The point resistance in sand continues to increase gradually. This is probably why a pile in sand does not usually reach a plunging failure during a load test.

The relation between the lateral resistance and displacement for a 36-in. diameter pile in clay and sand is shown in Figure 6.8.

6.6.3.5 Laterally loaded piles reaction

The pile foundation should be designed to sustain lateral loads, whether static or cyclic. Additionally, the designer should consider overload cases in which the design lateral loads on the platform foundation are increased by an appropriate safety factor.

The designer should be satisfied that the overall structural foundation system will not fail under overloads. The lateral resistance of the soil near the surface is significant to pile design, and the effects of scour and soil disturbance on this resistance during pile installation should be considered. Generally, under lateral loading, clay soils behave as a plastic material, which makes it necessary to relate pile-soil deformation to soil resistance. To facilitate this procedure, lateral soil resistance deflection (p - y) curves should be constructed using stress-strain data from laboratory soil samples. The ordinate for these curves is soil resistance, p , and the abscissa is soil deflection, y . By iterative procedures, a compatible set of load-deflection values for the pile-soil system can be developed.

Matlock (1970) performed a comprehensive study of the design of laterally loaded piles in soft clay, and Reese and Cox (1975) performed a study of laterally loaded piles in stiff clay.

It is noted that these p - y curves are recommended for estimating pile bending moment, displacement, and rotation profiles for various (static or cyclic) loads. Different criteria may be applicable for fatigue analysis of a pile that has previously been subjected to loads larger than those used in the fatigue analysis and that resulted in “gapping” around the top of the pile.

6.6.3.6 Lateral bearing capacity for soft clay

According to API RP2KA (2007), for static lateral loads, the ultimate unit lateral bearing capacity of soft clay p_u has been found to vary between $8c$ and $12c$, except at shallow depths, where failure occurs in a different mode due to minimum overburden pressure. Cyclic loads cause deterioration of lateral bearing capacity below that for static loads.

In the absence of more definitive criteria, the following is recommended: p_u increases from $3c$ to $9c$ as X increases from 0 to X_R according to

$$p_u = 3c + \gamma X + J(cX/D) \quad (6.17)$$

and

$$p_u = 9c, \text{ for } X \geq X_R \quad (6.18)$$

For a condition of constant strength with depth, equations (6.17) and (6.18) are solved simultaneously to give the following equation:

$$X_R = 6D/[(\gamma D/c) + J] \quad (6.19)$$

where p_u = ultimate resistance (in kPa), c = undrained shear strength for undisturbed clay soil samples (in kPa), D = pile diameter (in mm), γ = effective unit weight of soil (in MN/m³), J = a dimensionless empirical constant with values ranging from 0.25 to 0.5 (having been determined by field testing; a value of 0.5 is appropriate for Gulf of Mexico clays, so it should be checked in another areas), X = depth below soil surface (in mm), and X_R = depth below soil surface to bottom of reduced resistance zone (in mm).

Where the strength varies with depth, equations (6.19) and (6.20) may be solved by plotting the two equations, that is, p_u versus depth. The point of first intersection of the two equations is taken to be X_R .

These empirical relationships may not apply where strength variations are erratic. In general, minimum values of X_R should be about 2.5 pile diameters.

On the other hand, in soft clay, the load-deflection (p - y) curves for lateral soil resistance-deflection relationships for piles are generally nonlinear. The p - y curves

Table 6.12 Relation between pile lateral load and lateral deflection

p/p_u	y/y_c
0.00	0.00
0.23	0.1
0.33	0.3
0.5	1.0
0.72	3.0
1.00	8.00
1.00	∞

Table 6.13 Relation between pile load and lateral displacement

$X > X_R$		$X < X_R$	
p/p_u	y/y_c	p/p_u	y/y_c
0.00	0.00	0.00	0.0
0.23	0.1	0.23	0.1
0.33	0.3	0.33	0.3
0.50	1.0	0.50	1.0
0.72	3.0	0.72	3.1
0.72	∞	0.72 X/X_R	15.0
		0.72 X/X_R	∞

for the short-term static load case may be generated from Table 6.12, where p = actual lateral resistance (in kPa), y = actual lateral deflection (in m),

$$y_c = 2.5\varepsilon_c D \quad (6.20)$$

and ε_c is the strain that occurs at one half the maximum stress on laboratory unconsolidated, undrained, compression tests of undisturbed soil samples.

For the case where equilibrium has been reached under cyclic loading, the p - y curves may be generated from Table 6.13.

6.6.3.7 Lateral bearing capacity for stiff clay

For static lateral loads, the ultimate bearing capacity p_u of stiff clay ($c > r$ 96 kPa), as for soft clay, would vary between $8c$ and $12c$. Due to rapid deterioration under cyclic loading, the ultimate resistance is reduced to something considerably less and should be so considered in cyclic design. Furthermore, because stiff clays also have nonlinear stress-strain relationships, they are generally more brittle than soft clays. In developing stress-strain curves and subsequent p - y curves for cyclic loads,

good judgment should reflect the rapid deterioration of load capacity at large deflections for stiff clays.

In general, under lateral loads, clay soils behave as a plastic material, which makes it necessary to relate pile-soil deformation to soil resistance.

For a more detailed study of the construction of p - y curves, see Matlock (1970) for soft clay, Reese and Cox (1975) for stiff clay, O'Neill and Murchison (1983) for sand, and Georgiadis (1983) for layered soils.

6.6.3.8 Lateral bearing capacity for sand

A series of studies verified the theoretical studies with the field-test results during lateral loading of a 24-in. diameter test pile installed at sites with clean, fine sand, and silty sand. (The studies were funded by Amoco's production company, Chevron oil field research, Esso's production research company, Mobil Oil Corporation, and Shell's development company.) The results suggest a shape for the p - y curve as shown in Figure 6.8 where the initial part is a straight line representing the elastic behavior and the horizontal straight line represents the plastic behavior, with the straight lines connected by a parabola.

The values of P_m and p_c are a function of the ultimate soil resistance. A difference between the ultimate resistance from theory and that from experiments was observed, which was covered by empirical factors. Another study, by O'Neill and Murchison (1983), evaluated p - y relationships in sand. API RP2A (2007) recommends that the p - y curve be calculated using the information from that study.

The ultimate lateral bearing capacity for sand has been found to vary from a value at shallow depths determined by equation (6.21) to a value at deep depths determined by equation (6.22). At a given depth, the equation giving the smallest value of p_u should be used as the ultimate bearing capacity.

$$p_{us} = (C_1 H + C_2 D) \gamma H \quad (6.21)$$

$$p_{ud} = C_3 D \gamma H \quad (6.22)$$

where p_u = ultimate resistance (force/unit length) (in kN/m) (s = shallow, d = deep), γ = effective soil weight (in kN/m³), H = depth (in m), ϕ' = angle of internal friction of sand (in degrees), C_1 , C_2 , C_3 = coefficients determined from Table 6.14 as function of ϕ' , and D = average pile diameter from surface to depth (in m).

The relationship between lateral soil resistance and deflection (p - y curve) for sand is nonlinear. If no definitive information is available, the curve may be approximated at any specific depth H , according to API RP2A, by the following equations:

$$P = A p_u \tan h \left[\frac{kH}{A p_u} y \right] \quad (6.23)$$

where A is a factor to account for the cyclic or static loading condition, evaluated by $A = 0.9$ for cyclic loading and by $A \geq 0.9$ for static loading, so that

Table 6.14 Coefficients C_1 , C_2 , C_3

Angle of internal friction, ϕ	C_1	C_2	C_3
20	0.6	1.5	8.5
21	0.7	1.6	9.6
22	0.8	1.7	10.8
23	0.9	1.8	12.2
24	1.0	1.9	13.8
25	1.1	2.0	15.6
26	1.2	2.1	17.6
27	1.3	2.2	19.9
28	1.4	2.3	22.5
29	1.6	2.5	25.4
30	1.7	2.6	28.7
31	1.9	2.7	32.4
32	2.1	2.9	36.6
33	2.3	3.0	41.4
34	2.5	3.2	46.7
35	2.8	3.4	52.8
36	3.1	3.6	59.6
37	3.4	3.8	67.4
38	3.8	4.0	76.1
39	4.2	4.2	86.0
40	4.6	4.4	101.5

$$A = [3.0 - 0.8(H/D)] \quad (6.24)$$

where p_u = ultimate bearing capacity at depth H (in kN/m); k = initial modulus of subgrade reaction (in kN/m³), as determined from [Table 6.15](#) as function of the angle of internal friction, ϕ' , and the relative density for sand under the water table. y = lateral deflection (in inches or m); and H = depth (in m).

6.6.3.9 Changes in axial capacity in clay with time

The pile capacity calculated from the previous equation does not consider the effect of time on the pile capacity. Note that, in the old platform constructed 40 years ago and more, if you review the calculation, you find that its factor of safety does not follow that proposed by the API and the environment over time will surely affect the pile capacity. Normal phenomena after longtime there will be a good bond between the pile and the surrounding soil and this additional adhesion is not considered in the calculation. Therefore, a studies were performed to define the behavior of the axial capacity in clay soil with time. [Clarke \(1993\)](#) and [Bogard and Matlock \(1990\)](#)) conducted field measurements studies in which it was shown that the time

Table 6.15 Relation between subgrade reaction, angle of internal friction, and relative density for sand below the water table

Soil type	Angle of internal friction, ϕ	Relative density, %	Subgrade reaction, K , t/m^3
Very loose Loose sand	<29	20	265.7
		25	426.3
		30	553.6
		35	744.6
Medium dense	29–30	40	996.4
		45	1356.3
		50	1716.1
		55	2026.1
Dense	30–36	60	2491.1
		65	2850.9
		70	3293.8
		75	3792.0
		80	4262.6

required for driven piles to reach ultimate capacity in a cohesive soil can be relatively long, as much as 2–3 years.

It is worth mentioning that the rate of strength gain is highest immediately after driving, and the rate decreases during the dissipation process. Therefore, a significant strength increase can occur in a relatively short time.

During pile driving in normally to lightly overconsolidated clay, the soil surrounding a pile is significantly disturbed, the stress state is altered, and large excess pore pressures can be generated. After installation, these excess pore pressures begin to dissipate, which means that the soil mass around the piles begins to consolidate, so the pile capacity increases with time. This process is usually referred to as *setup*. The rate of excess pore pressure dissipation is a function of the coefficient of radial consolidation, pile radius, plug characteristics, and soil layering.

In the most popular case, where the driven pipe piles supporting a structure have design loads applied to the piles shortly after installation, the time-consolidation characteristics should be considered in pile design. Noting that, in traditional, fixed offshore structure installations, the time between the pile installation and the platform being total loaded is in the range of 1–3 months. In some cases, the commissioning and startup come early; in this case, such information should be transferred to the engineering office, as the expected increase in capacity with time is an important design variable that can affect the safety of the foundation system during early stages of the consolidation process.

The relation between the pore pressure and load test data at different times after pile driving is expressed by an empirical correlation: the degree of consolidation, degree of plugging, and pile shaft shear transfer capacity. The test results for closed-ended steel piles in heavily overconsolidated clay indicate no significant

change in capacity with time. This is contrary to tests on 0.273 m (10.75 in.) diameter closed-ended steel piles in overconsolidated Beaumont clay, where considerable and rapid setup in 4 days was found, so the pile capacity at the end of installation was never fully recovered.

So, it is very important to highlight that the axial capacity of the pile over time is under research and development, and there is no solid formula or equation to follow. Focus is on the research done on the specific site locations and depends on the previous history of this location.

6.6.4 Pile capacity calculation methods

API RP2A (2007) presents new methods for calculating pile capacity based on the cone penetration test.

As it is presented previously, a simple method for assessing pile capacity in cohesionless soils, this method is recommended in previous editions of API RP2A-WSD. Changes were made to remove potential nonconservatism in previous editions. There are reliable CPT-based methods for predicting pile capacity. All these methods are based on direct correlations of pile unit friction and end-bearing data with cone tip resistance (q_c) values from CPT. These CPT-based methods cover a wider range of cohesionless soils, are considered fundamentally better, and have shown statistically closer predictions of pile load test results.

The new CPT-based methods for assessing pile capacity in sand are preferred to the previous method. However, more experience is required with all these new methods before any single one can be recommended for routine design over previously presented methods. API states clearly that the new CPT-based methods should be used only by qualified engineers who are experienced in interpreting CPT data and who understand the limitations and reliability of the CPT-based methods.

The assumption is made that friction and end-bearing components are uncoupled. Hence, for all methods, the ultimate bearing capacity in compression (Q_d) and tensile capacity (Q_t) of plugged open-ended piles are determined by

$$Q_t = Q_f + Q_p = P_o \int f_{c,z} dz + A_q q_p \quad (6.25)$$

$$Q_f = P_o \int f_{t,z} dz \quad (6.26)$$

Note that, since the friction component, Q_f , involves numerical integration, results are sensitive to the depth increment used, particularly for CPT-based methods. As guidance, depth increments for CPT-based methods should be on the order of $\{1/100\}$ of the pile length (or smaller). In any case, the depth increment should not exceed 0.3 m.

Four recommended CPT-based methods are mentioned in API RP2A:

1. Simplified ICP-05.
2. Offshore UWA-05 (Lehane et al., 2005a,b).
3. Fugro-05 (Lehane et al., 2005a; Kolk et al., 2005).
4. NGI-05 (Lehane et al., 2005a; Clausen et al., 2005).

The first method is a simplified version of the design method recommended by Jardine et al. (2005), whereas the second is a simplified version of the UWA-05 method applicable to offshore pipe piles. Methods 2, 3 and 4 are summarized by Lehane et al. (2005a). Friction and end-bearing components should not be taken from different methods.

The unit skin friction formulas for open-ended steel pipe piles for the first three recommended CPT-based methods (Simplified ICP-05, Offshore UWA-05, and Fugro-05) can be considered as special cases of the general formula

$$f_z = u \cdot q_{cz} \left(\frac{\sigma'_{vo}}{P_a} \right) A_r^b \left[\max \left(\frac{L-z}{D}, v \right) \right]^{-c} [\tan \delta_{cv}]^d \left[\min \left(\frac{L-z}{D}, 1 \right) \right] \quad (6.27)$$

where f_z is the unit skin friction, δ_{cv} is pile–soil constant-volume interface friction angle, L is the pile length underneath the seabed, $A_r = 1 - (D_i/D)^2$, D_i is the pile inner diameter ($D_i = D - 2t$), z is the depth under the seabed, q_{cz} is the CPT tip resistance at depth z , D is the outer diameter, t is the wall thickness, and P_a is the atmospheric pressure equal to 100 kPa.

Table 6.16 provides the recommended values for parameters a , b , c , d , e , u , and v for compression and tension, which are the unit skin friction parameter values for driven open-ended steel piles for the Simplified ICP-05, Offshore UWA-05, and Fugro-05 methods.

Additional recommendations for computing unit friction and end bearing for all four CPT-based methods are presented in the following subsections.

Table 6.16 Unit skin friction parameter values for driven open-ended steel pipes (simplified ICP-05, Offshore UWA-05, and Fugro-05 methods)

Method	Load direction	a	b	c	d	e	u	v
Simplified ICP-05	Compression	0.1	0.2	0.4	1	0	0.023	$4\sqrt{A_r}$
	Tension	0.1	0.2	0.4	1	0	0.016	$4\sqrt{A_r}$
Offshore UWA-05	Compression	0	0.3	0.5	1	0	0.030	2
	Tension	0	0.3	0.5	1	0	0.022	2
Fugro-05	Compression	0.05	0.45	0.90	0	1	0.043	$2\sqrt{A_r}$
	Tension	0.15	0.42	0.85	0	0	0.025	$2\sqrt{A_r}$

6.6.4.1 Application of CPT

By using CPT measurement as the basis for the previous methods to calculate the unit skin friction and end bearing for the pile, some precautions and information should be considered as in obtaining t - z data for an axial load-deformation response. The peak unit skin friction in compression and tension at a given depth, f_{cz} and f_{tz} , are not unique and both depend on the pile geometry. In general, the axial load and deformation response are affected by the pile penetration depth, the pile diameter, and its wall thickness. Note that an increased pile penetration decreases these ultimate values at a given depth.

In doing the test to obtain the q - z data for an axial load-deformation response, the end bearing (Q_p) is assumed to be fully mobilized at a pile tip-displacement value of $0.1D$.

Soil types such as carbonate sands, micaceous sands, glauconitic sands, volcanic sands, silts, and clayey sands have unusually weak structures with compressible grains. These types require special consideration in-situ and laboratory tests for selection of an appropriate design method and design parameters, according to [Thompson and Jardine \(1998\)](#) and [Kolk \(2000\)](#) for pile design in carbonate sand and to [Jardine et al. \(2005\)](#).

It is worth to mention that using CPT in cohesionless soil, such as gravels when particle sizes are in excess of 10% of the CPT cone diameter, is misleading, and one possible approach could be to use the lower-bound q_c profile. In this case, one can estimate the end-bearing capacity profile from the adjacent sand layers.

In using CPT in weaker clay layers near the pile tip, it is recommended that q_c data averaged between $1.5D$ above the pile tip to $1.5D$ below the pile tip level should generally be satisfactory, provided q_c does not vary significantly. The UWA method should be used if significant q_c variations occur to compute $q_{c,av}$.

Using CPT data on a thin clay layer, which is less than around $0.1 D$ thick, is a problem especially when CPT data are discontinuous vertically or not all pile locations have been investigated. From practical point of view, the offshore piles usually develop only a small percentage of q_p under extreme loading conditions. So, the finite element method can be used in calculating the pile capacity, and settlement of a pile tip on sand containing weaker layers may be considered to assess the axial pile response under such conditions.

It is recommended that the end-bearing component be reduced if the pile tip is within a zone up to $\pm 3D$ from such layers. When q_c data averaging is also applied to this $\pm 3D$ zone, the combined effects may be unduly cautious, and such results should be critically reviewed. This rule applies also in cases of large pile diameter ($D > 2$ m).

6.7 Pile wall thickness

The wall thickness of the pile may vary along its length and may be controlled at a particular point by any one of several loading conditions or requirements. The minimum wall thickness is shown in [Table 6.17](#).

Table 6.17 Minimum pile wall thickness

Pile diameter, in mm (in.)	Nominal wall thickness, in mm (in.)
610 (24)	13 ($\frac{1}{2}$ ")
762 (30)	14 ($\frac{5}{16}$ ")
914 (36)	16 ($\frac{3}{8}$ ")
1067 (42)	17 ($\frac{11}{16}$ ")
1219 (48)	19 ($\frac{3}{4}$ ")
1524 (60)	22 ($\frac{7}{8}$ ")
1829 (72)	25 (1")
2134 (84)	28 ($1\frac{1}{8}$ ")
2438 (96)	31($1\frac{1}{4}$ ")
2743 (108)	34 ($1\frac{3}{8}$ ")
3048 (120)	37 ($1\frac{1}{2}$ ")

The allowable pile stresses should be the same as those permitted by the American Institute of Steel Construction (AISC) specification for a compact hot-rolled sections. A rational analysis considering the restraints placed on the pile by the structure and the soil should be used to determine the allowable stresses for the portion of the pile that is not laterally restrained by the soil. General column buckling of the portion of pile below the mudline need not be considered unless the pile is believed to be laterally unsupported because of extremely low soil shear strengths, large computed lateral deflections, or some other reason.

6.7.1 Design pile stresses

The pile wall thickness in the vicinity of the mudline, and possibly at other points, is normally controlled by the combined axial load and bending moment that result from the design loading conditions for the platform.

The moment curve for the pile may be computed with soil reactions determined taking into consideration possible soil removal by scour. It may be assumed that the axial load is removed from the pile by the soil at a rate equal to the ultimate soil-pile adhesion divided by the appropriate pile safety factor, as specified in [Table 6.9](#). When lateral deflections associated with cyclic loads at or near the mudline are relatively large, exceeding y_c for soft clay, consideration should be given to reducing or neglecting the soil-pile adhesion through this zone.

6.7.2 Stresses due to the weight of the hammer during hammer placement

Each pile or conductor section on which a pile hammer will be placed should be checked for stresses due to placing the equipment. These loads may be the limiting factors in establishing the maximum length of add-on sections. This is particularly true in cases where piling will be driven or drilled on a batter. The most frequent

effects include static bending, axial loads, and arresting lateral loads generated during initial hammer placement.

Experience indicates that reasonable protection from failure of the pile wall due to these loads is provided if the static stresses are calculated as follows:

1. The pile projecting section should be considered a freestanding column with a minimum effective length factor, K , of 2.1 and a minimum reduction factor, C_m , of 1.0.
2. Bending moments and axial loads should be calculated using the full weight of the pile hammer, cap, and leads acting through the center of gravity of their combined masses, and the weight of the pile add-on section, with due consideration to pile batter eccentricities.

The bending moment so determined should not be less than that corresponding to a load equal to 2% of the combined weight of the hammer, cap, and leads applied at the pile head and perpendicular to its centerline.

Allowable stresses in the pile should be calculated with the allowable stress design by AISC. Note that the one-third increase in stress should not be allowed.

Consideration should also be given to the stresses that occur in the freestanding pile section during driving, as shown in Figures 6.9 and 6.10. Generally, stresses

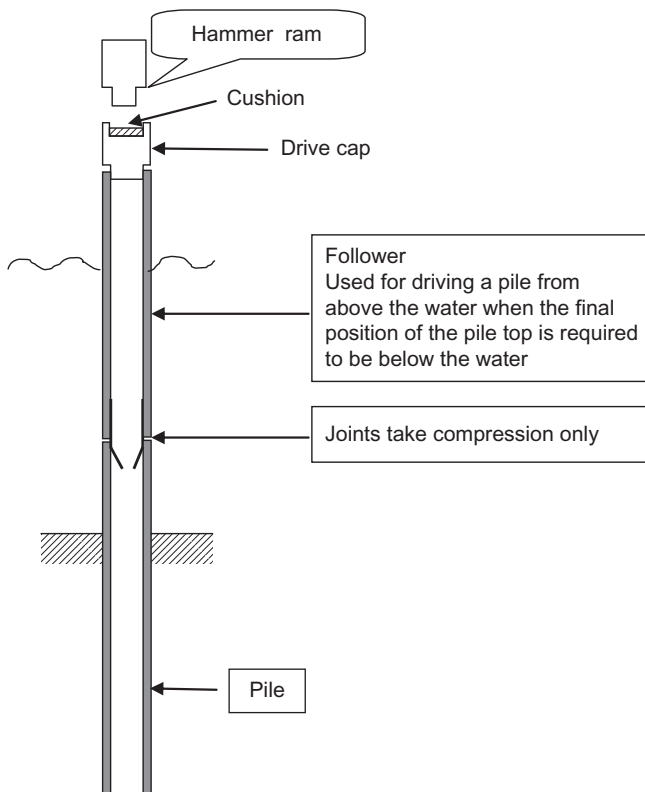


Figure 6.9 Typical arrangement for pile driving.

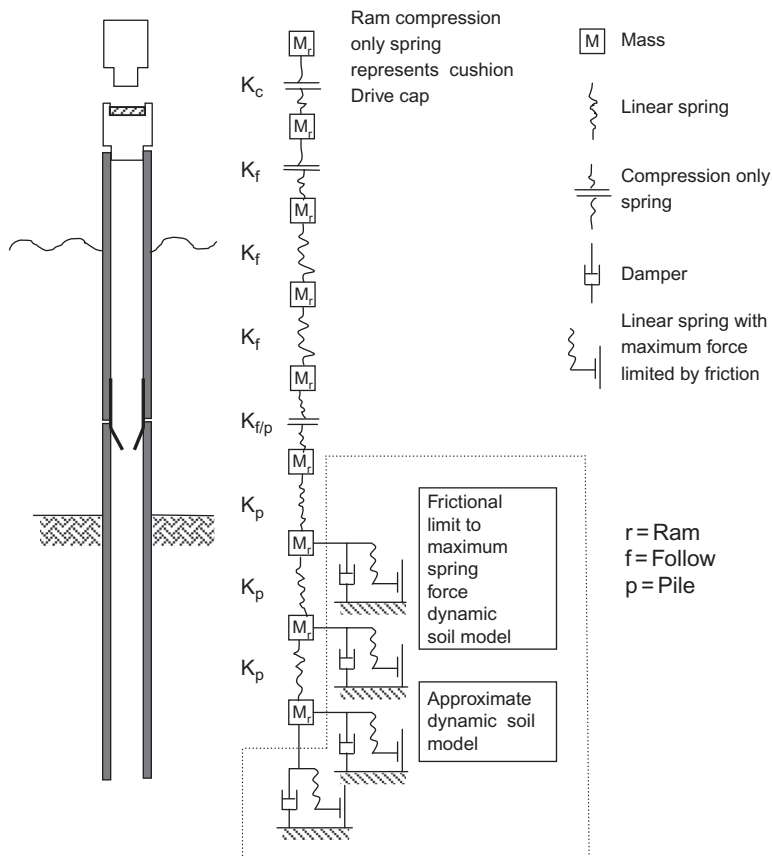


Figure 6.10 Dynamic analysis model of pile driving.

are checked based on the conservative criterion that the sum of the stresses due to the dynamic stresses caused by the impact from the hammer and the stresses due to axial load and bending (the static stresses) should not exceed the minimum yield stress of the steel.

Less conservative criteria are permitted, provided that they are supported by sound engineering analysis and empirical evidence. A method of analysis based on wave propagation theory should be used to determine the dynamic stresses.

In general, it may be assumed that column buckling will not occur as a result of the dynamic portion of the driving stresses. The dynamic stresses should be less than 80–90% of yield, depending on specific circumstances, such as the location of the maximum stresses down the length of pile, the number of blows, previous experience with the pile-hammer combination, and the confidence level in the analyses. Separate considerations apply when significant driving stresses may be transmitted into the structure and damage to appurtenances must be avoided.

The static stress during driving may be taken to be the stress resulting from the weight of the pile above the point of evaluation plus the pile hammer components actually supported by the pile during the hammer blows, including any bending stresses resulting from these. With hydraulic hammers, it is possible that the driving energy may exceed the rated energy, and this should be considered in the analyses. Also, the static stresses induced by hydraulic hammers need to be computed with special care, due to the possible variations in driving configurations, for example, when driving vertical piles without lateral restraint and exposed to environmental forces. The pile hammers evaluated for use during driving should be noted by the designer on the installation drawings or specifications.

In the past, several case histories were reported that describe some of the unusual characteristics of foundations on carbonate soils and their often poor performance. It has been shown from numerous pile load tests that piles driven into weakly cemented and compressible carbonate soils mobilize only a fraction of the capacity, as low as 15%, predicted by conventional design/prediction methods for siliceous material of the type generally encountered in the Gulf of Mexico. On the other hand, dense, strongly cemented carbonate deposits can be very competent foundation material. Unfortunately, the difficulty in obtaining high-quality samples and the lack of generalized design methods sometimes make it difficult to predict where problems may occur.

The energy is determined primarily by the mass of the ram and its impact velocity:

$$E = 0.5 mV^2 \quad (6.28)$$

Note that only 60–70% of the energy is typically transferred to the drive cap from the ram. It is obvious that the greater the energy, the greater will be the penetration; on the other hand, the greater will be the risk of damaging the pile. The maximum stress in the stress wave is largely determined by the velocity of the ram. It is worth mentioning that, for easy driving conditions, long duration and a low stress waveforms are the best and this could be achieved by a heavier and slower ram and a soft cushion.

Using the finite difference method, the ram is represented by a concentrated mass, and the required information about the ram is available from the hammer manufacturer (Table 6.18). The efficiency of the hammer depends on the conditions as well as the operating procedures, as shown in Table 6.19 for various hammer types.

6.7.3 Minimum wall thickness

API RP2A defines the minimum wall thickness of the pile based on the D/t ratio of the entire length of a pile should be small enough to preclude local buckling at stresses up to the yield strength of the pile material. Consideration should be given to the different loading situations occurring during the installation and service life of a piling. For in-service conditions and those installations where normal pile driving is anticipated or where piling installation will be by means other than driving, the limitations should be considered to be the minimum requirements.

Table 6.18 Pile hammering data

Contract no.:	Project:	Structure name:	
Pile driving contractor			
Hammer	Manufacturer:	Model No.	
	Hammer type:	Serial no.	
	Manufacturers maximum rated energy, m-KN	Stroke at maximum rated energy, m	Range in operating energy to m-kN
	Range in operating stroke to m-kN	Rame weight, kips	
Strike plate	Weight, kg	Diameter mm	Thickness mm
Hammer cushion	Area, mm ²	Plate thickness, mm	
	No. of plates, mm	Total thickness of hammer cushion, mm	
Helmet (drive head)	Weight, kg		
Pile cushion	Materials:	Area, mm ²	
	Sheet thickness, mm	No. of sheet:	
	Total thickness of pile cushion, mm		
Pile	Pile type:	Wall thickness, mm	
	Cross-sectional area: mm ²	Weight/m:	
	Ordered length: m	Design load, ton	
	Ultimate pile capacity, ton		

Table 6.19 Efficiency for different hammer types

Hammer	Efficiency
Single-acting steam or air	0.75–0.85
Double-acting steam or air	0.70–0.80
Diesel	0.85–1.0
Hydraulic	0.85–0.95

For piles that are to be installed by driving where sustained hard driving (250 blows per foot, 820 blows per meter) with the largest size hammer is anticipated to be used, the minimum piling wall thickness used should be more than

$$t = 6.35 + D/100 \quad (6.29)$$

where t = wall thickness (in in. or mm) and D = diameter (in in. or mm).

The preceding requirement for a smaller D/t ratio when hard driving is expected may be relaxed when it can be shown by past experience or by detailed analysis that the pile will not be damaged during its installation.

6.7.4 Driving shoe and head

The purpose of driving shoes is to assist piles to penetrate through hard layers or to reduce driving resistances, allowing greater penetrations to be achieved than would otherwise be the case. Different design considerations apply for each use.

If an internal driving shoe is provided to drive through a hard layer, it should be designed to ensure that unacceptably high driving stresses do not occur at and above the transition point between the normal and the thickened section at the pile tip. Also, it is important to check that the shoe does not reduce the end-bearing capacity of the soil plug below the value assumed in the design. External shoes are not normally used, as they tend to reduce the skin friction along the length of pile above them.

The installation contractor is responsible for designing the driving head at the top of the pile, and it should be designed to ensure that it is fully compatible with the proposed installation procedures and equipment.

6.7.5 Pile section lengths

In defining the pile section lengths, consideration should be given to the following:

1. The capability of the lift equipment to raise, lower, and stab the sections.
2. The capability of the lift equipment to place the pile-driving hammer on the sections to be driven.
3. The possibility of a large amount of downward pile movement immediately following the penetration of a jacket leg closure.
4. Stresses developed in the pile section while lifting.
5. The wall thickness and material properties at field welds.
6. Avoiding interference with the planned concurrent driving of neighboring piles.
7. The type of soil in which the pile tip is positioned during driving interruptions for field welding to attach additional sections. In addition, static and dynamic stresses due to the hammer weight and operation should be considered. Each pile section on which driving is required should contain a cutoff allowance to permit the removal of material damaged by the impact of the pile-driving hammer. The normal allowance is 2 to 5 ft (0.5 to 1.5 m) per section. Where possible, the cut for the removal of the cutoff allowance should be made at a conveniently accessible elevation.

6.8 Pile drivability analysis

The pile drivability analysis has three stages:

1. Evaluation of soil resistance to driving.
2. Wave equation analysis.
3. Estimate of blow count versus pile penetration.

The procedures used to evaluate the soil resistance to driving are empirical and have been developed from the back-analysis of pile-driving records. Their use is therefore limited to pile drivability assessment by wave equation analysis and they are not intended to provide an estimate of the ultimate axial capacity of foundation piles.

In the present case, as is often true, some of the parameters required for the wave equation analysis (step 2) depend on the maximum achievable penetration, which is calculated in step 3. Therefore, to ensure consistency between the steps, an iterative analysis has been carried out.

6.8.1 Evaluation of soil resistance drive

Different procedures are used for evaluating the soil resistance drive (SRD) in cohesionless and cohesive soils. They are discussed next. As is the case for static pile capacity analysis, the components of shaft resistance and end bearing in SRD are evaluated separately, then combined to give total driving resistance (Toolan and Fox, 1977).

The variability of the soil conditions across the site and some anticipated variation in hammer performance are likely to influence the apparent driving resistance. Furthermore, the driving resistance during continuous driving is known to be considerably lower than when driving is restarted after an interruption, long enough to allow soil setup. To account for these factors, upper-bound and lower-bound SRD profiles have been formulated for a given design soil profile, based on the recommendations of Stevens et al. (1982).

6.8.2 Unit shaft resistance and unit end bearing for uncemented materials

In cohesive soil, the unit skin friction has been assessed based on the method proposed by Simple and Gemeinhardt (1981). This method was developed from back-analysis of pile installations in normally consolidated to heavily overconsolidated clays from many areas.

The unit shaft resistance component of SRD is derived using the API RP (1984) procedure for static pile capacity, modified by a pile capacity factor F_p , which is a function of the overconsolidation ratio (OCR), as follows:

$$F_p = 0.5(\text{OCR})^{0.3} \quad (6.30)$$

The unit end-bearing component of SRD is taken as $9S_u$.

The OCR is defined as the ratio of the maximum past effective consolidation stress and the present effective overburden stress. OCR is a function for undrain shear strength ratio (S_u/p') which is equal to 0.22 in normally consolidated clay with shear stress angle equal to 26° . In general, OCR is obtained by CPT.

In cohesionless (granular) soil, shaft resistance and end-bearing components of SRD can be derived using the API (1984) procedure for static pile capacity together with the soil parameters specified by [Stevens et al. \(1982\)](#) for the particular soil type. A limiting skin friction of 15 kPa has been taken for the calcareous and carbonate sand layers.

6.8.3 Upper- and lower-bound SRD

Based on [Stevens et al. \(1982\)](#), four cases have been assessed to obtain upper- and lower-bound SRD values in uncemented and weakly cemented soil layers:

1. SRD for lower-bound, coring pile is the outside shaft friction in addition to the inside shaft friction as half the outside. End bearing is unit end bearing multiplied by the steel annulus area.
2. SRD for upper-bound, coring pile is similar to the previous case but with the full shaft friction on the inside of the pile.
3. SRD for lower-bound, plugged pile is the outside shaft friction in addition to the end bearing multiplied by full cross-section area.
4. SRD for upper-bound, plugged pile for cohesionless soil layers is with outside shaft friction from case 3 increased by 30% and end bearing from case 3 increased by 50%. For cohesive layers, the outside shaft friction is as in case 3 and end bearing from case 3 is increased by 67%.

For the sandstone and limestone layers, plugged conditions are unrealistic, and only the coring cases are considered.

The software GRLEAP input data and the method of obtaining the compressive strength on the pile and the number of blows every 250 mm is presented in [Figure 6.11](#).

SRD calculations have been made to cover the range in unit values just set out. Sample results are shown in [Figure 6.12](#). As shown in the figure, a range in SRD is provided, reflecting the various combinations in unit values given previously. The lower bounds are likely to be applicable to the minimum resistances during continuous driving, with upper bounds indicative of local variations in soil conditions and resistances expected immediately on restarting a drive (i.e., soil setup condition).

Example 6.4

[Figure 6.11](#) presents the output from the GRLWEAP software by choosing the hammer type and pile length and its penetration. From the output curve, you have the load in the pile about 1480 kN.

Solution As per the chart of the analysis for hammer type Delmag D12-42,

Efficiency = 0.80

Helmet = 7.60 kN

Hammer cushion = 10,535 kN/mm

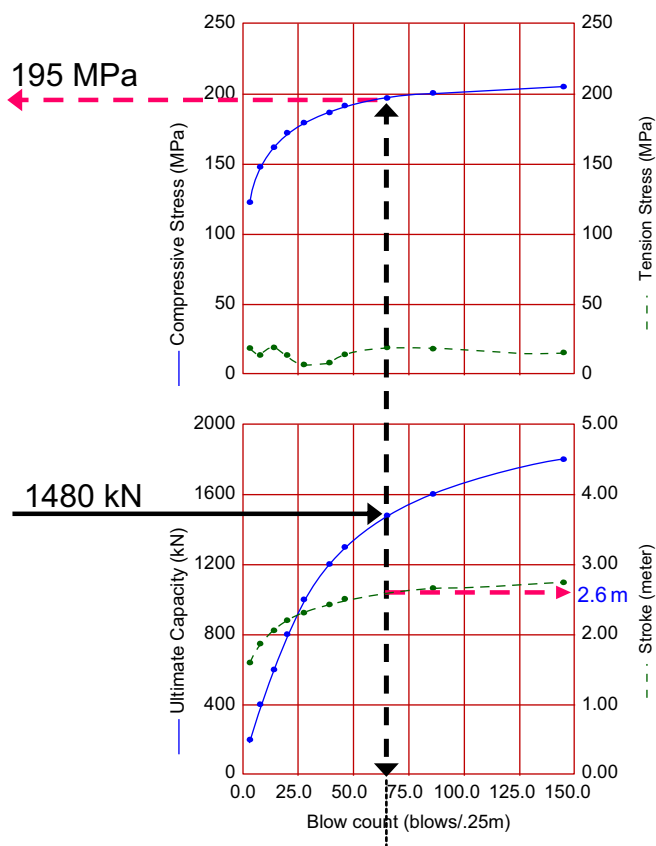


Figure 6.11 GRLWEAP example.

Based on the number of counts is around 60 counts for 250 mm depth and the compressive stress on the pile is 195 MPa, which should be check against the pile capacity.

Toe damping = 0.5 s/m
 Skin quake = 2.5 mm
 Toe quake = 3 mm
 Skin damping = 016 s/m
 Pile length = 25 m

6.8.4 Results of wave equation analysis

Blow count versus SRD curves have been developed for the hammers listed in Table 6.20, using a software program.

Achievable penetration is based on refusal criteria of 15 blows/0.25 m.

The input parameters for the wave equation analysis differed for each hammer, pile, and SRD bound, primarily in terms of the percentage of SRD that was due to

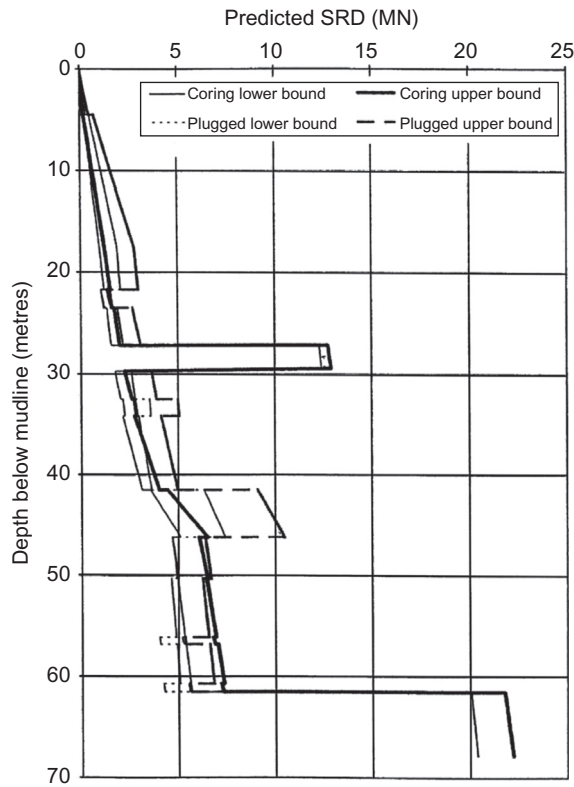


Figure 6.12 Present example for soil resistance to driving fora pipe pile 30" with 1¼" thickness.

Table 6.20 Summary of drivability analysis for pile 30 in. in diameter

Hammer	Achievable penetration	Maximum driving stresses
Delmag D-80	27.2	202
MRBS 3000	27.2	220
MRBS 3900	61.5	252

skin friction. This has an effect on the amount of energy lost through damping, because of differences in damping values for skin friction and end bearing. The following damping values were used:

- Skin damping for purely cohesive soil, 0.65 s/m.
- Skin damping for cohesionless soil, 0.15 s/m.
- Tip damping, all soils, 0.50 s/m.

Skin damping for a combination of soil types was assessed by taking account of the fraction of total skin resistance due to each type and linearly interpolating

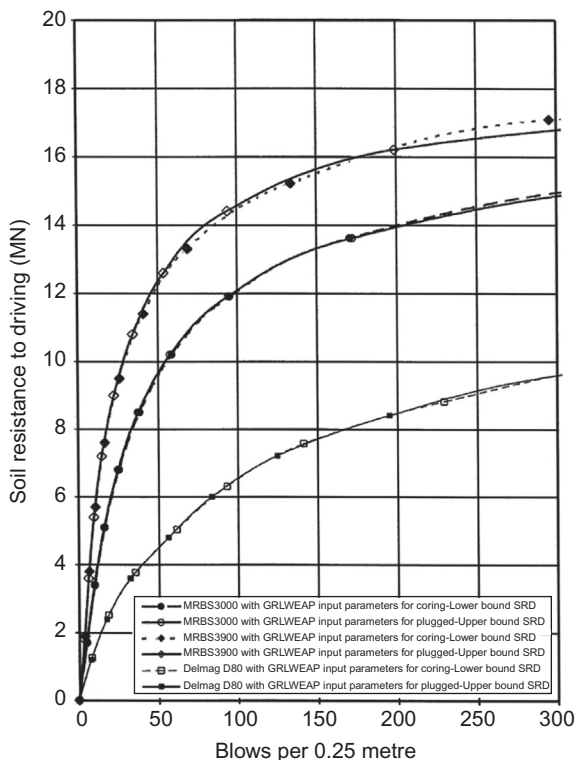


Figure 6.13 Present example for blow count versus SRD for 30" pipe pile with 1¼" wall thickness with three different hammers, 3000, 3900, and D80, Delmag.

between the purely cohesive and purely cohesionless values. Quakes of 0.1 in. were taken in all calculations.

The results are shown in [Figure 6.13](#). There are two curves for each hammer analyzed, corresponding to input parameters associated with the relevant lower- and upper-bound SRD cases.

6.8.5 Results of drivability calculations

By combining the SRD versus depth curves and the wave equation results, one obtains the predicted blow count versus depth. The results are shown in [Figure 6.14](#) and are summarized in [Table 6.20](#).

According to a case study in the Red Sea, refusal will occur at 27.2 m penetration if the Delmag 080 hammer is used. For the Menck MRBS 3000, high blow counts will be experienced during penetration and refusal may occur, for example, if the thickness or strength of cemented material at 27.2 m increases significantly from that identified in the borehole. For the Menck MRBS 3900, and indeed for all the hammers analyzed, refusal is indicated at 61.5 m penetration.

6.8.6 Recommendations for pile installation

The lower-bound curve of blow counts versus depth in Figure 6.14 applies if there is no interruption during driving. The upper-bound curves represent estimates of effects of delays during driving. To ensure an efficient offshore pile-driving operation, it is recommended that delays during driving be avoided if possible. Attention should be paid to ensuring that the blow count does not become excessive and that no pile-tip damage occurs, for all hammers analyzed. Depending on the pile-driving plant finally selected, it may be advisable to have equipment readily available during piling operations in case refusal or particularly high blow counts occur at 27.2 m below the mudline. If refusal occurs, an assessment may be needed of the effect on ultimate pile capacity and therefore target penetration.

The pile analysis usually assumes uniform wall thickness over the entire pile length. Should the wall thicknesses of the piles finally selected differ from this, then the drivability analysis should be repeated to assess the effect on blow counts and achievable penetration. It is recommended that consideration be given to

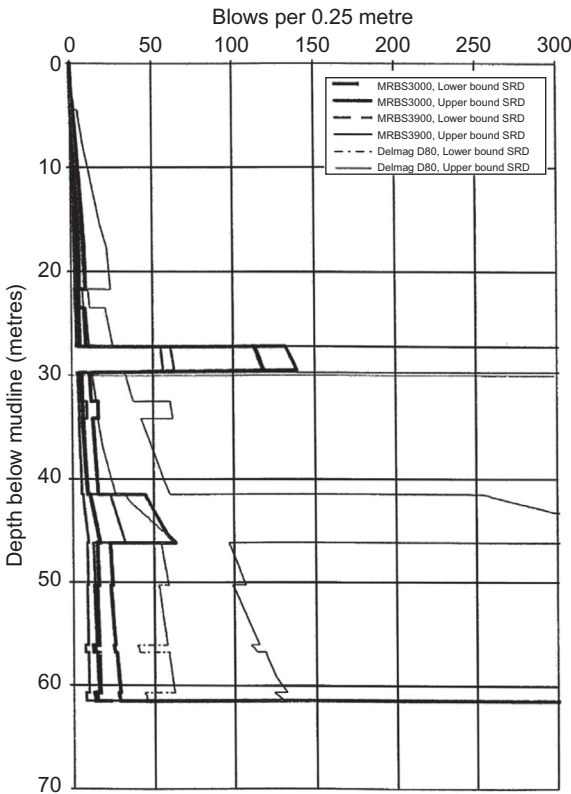


Figure 6.14 Presents example for blow count versus depth for three type of hammers for pile 30" with 1¼" wall thickness.

incorporation of a pile shoe to prevent undue distress to the pile during driving through the cemented layers.

The shoe should be externally flush, should have an appropriate outer bevel profile at the tip, and should have an increased wall thickness over a suitable length. As with an overall increase in pile wall thickness requiring some additional drivability assessment, the incorporation of a pile shoe also needs to be readdressed with regard to its effect on pile-driving behavior.

6.9 Soil investigation report

After all soil investigation tests on site and in the laboratory are completed, the following curves, which assist in pile design, should be presented in the soil investigation report: the curve for the relationship between the depth below the mudline and the unit skin friction ([Figure 6.15](#)) and the curve for the relationship between the pile depth below the mudline and the unit of end bearing ([Figure 6.16](#)).

The relation between the tension and compression pile capacity in different depth is as shown in [Figure 6.17](#).

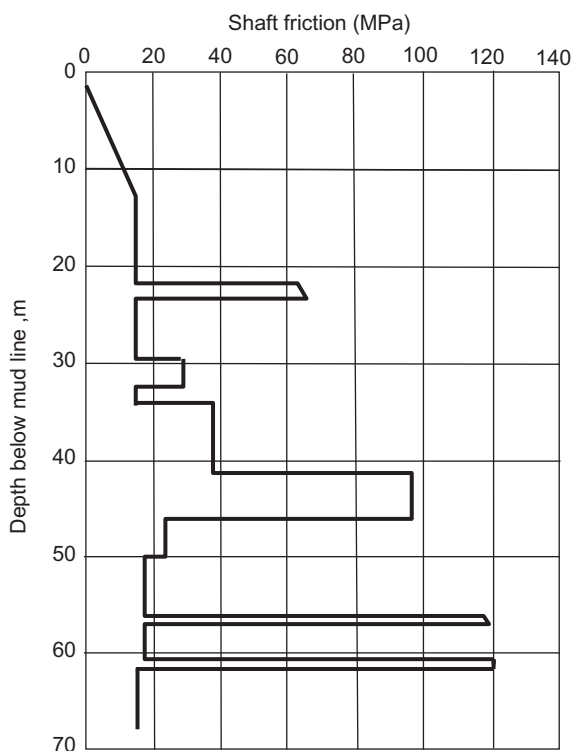


Figure 6.15 Relationship between the shaft friction capacity and the depth below the mudline.

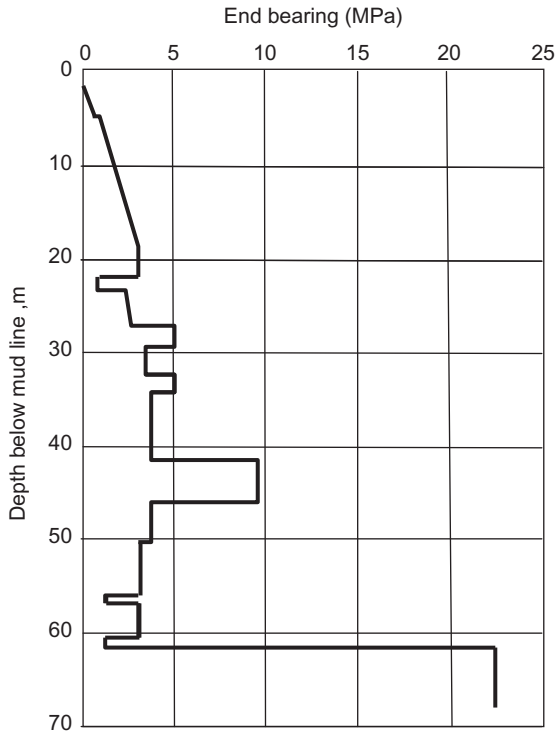


Figure 6.16 Relationship between the end-bearing capacity and the depth below the mudline.

If the pile diameter is not identified, the study should include the pile capacity in compression for different pile diameters, as presented in [Figure 6.18](#), and the pile capacity in tension for different pile diameters, as shown in [Figure 6.19](#).

Example 6.5

For a pile with a diameter of 1.22 m (48 in) and a depth of 80 m, the structure analysis by SACS present the compression axial force is equal to 8.2 MN in case of operating condition and 11 MN in case of extreme condition and the maximum tension force is 9 MN.

Solution As in [Figure 6.18](#), the compression capacity is 19.6 MN and the tension capacity, as shown in [Figure 6.19](#), is 13.8 MN.

To calculate the factor of safety under extreme conditions = $19.6/11 = 1.78 > 1.5$ OK.

To calculate factor of safety under operating conditions = $19.6/8.2 = 2.39 > 2.0$ OK.

To calculate the factor of safety under pile pullout = $13.8/9 = 1.53 > 1.5$ OK.

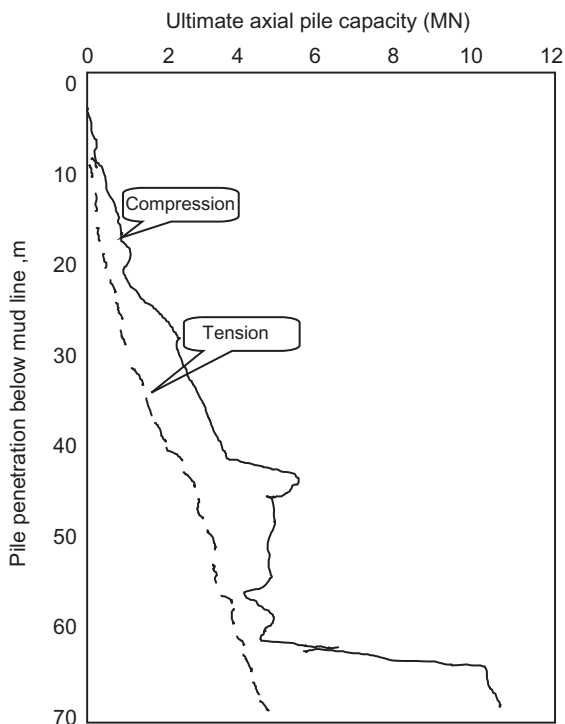


Figure 6.17 The ultimate axial pile capacity in tension and compression.

Example 6.6

For a pile with a diameter of 1.22 m (48 in) and a depth 80 m, the structure analysis by SACS presents the compression axial force as equal to 8.2 MN under operating conditions, 11 MN under extreme conditions, and the maximum tension force as 9 MN.

Solution As in Figure 6.18, the compression capacity is 19.6 MN and the tension capacity, as shown in Figure 6.19, is 13.8 MN.

To calculate the factor of safety under extreme conditions = $19.6/11 = 1.78 > 1.5$ OK.

To calculate factor of safety under operating conditions = $19.6/8.2 = 2.39 > 2.0$ OK.

To calculate the factor of safety under pile pullout = $13.8/9 = 1.53 > 1.5$ OK.

6.10 Composite pile

Grouting of piles platform loads may be transferred to steel piles by grouting the annulus between the jacket leg (or sleeve) and the pile. The load is transferred to the pile from the structure across the grout. Experimental work indicates that the mechanism of load transfer is a combination of bond and confinement friction between the grout, the steel surfaces, and the bearing of the grout against

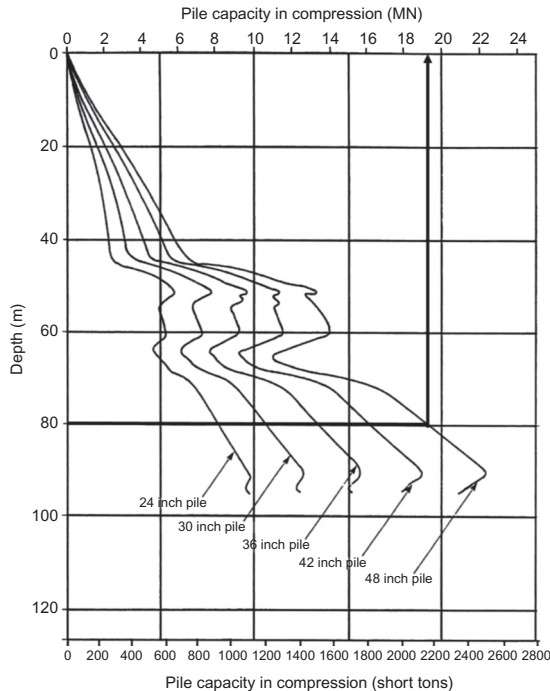


Figure 6.18 Relationship between the pile capacity in compression and depth by different pile diameters.

mechanical aids, such as shear keys. Centralizers should be used to maintain a uniform annulus or space between the pile and the surrounding structure. A minimum annulus width of $1\frac{1}{2}$ in. (38 mm) should be provided where grout is the only means of load transfer. Adequate clearance between pile and sleeve should be provided, taking into account the shear keys' outer dimension. Note that packers should be used as necessary to confine the grout. Proper means for the introduction of grout into the annulus should be provided, to minimize the possibility of dilution of the grout or formation of voids in the grout. The use of wipers or other means of minimizing mud intrusion into the spaces to be occupied by piles should be considered at sites having soft mud bottoms.

Grout is extensively used to “cement” the annulus between pile leg and jacket sleeve. An annular gap of 50–100 mm is usually selected. The grout should flow from the bottom up.

The mix is generally cement plus water. Fly ash may be used to replace part of the cement in order to reduce heat of hydration. Silica fume may be added to promote thixotropic behavior, increase strength, and reduce bleed. Admixtures may be used to provide water reduction, retardation, and expansion characteristics. It is important that trial batches be made to ensure that the grout has the proper flow characteristics as well as strength. Flow rate should be kept low to avoid entrapment of voids. Grout

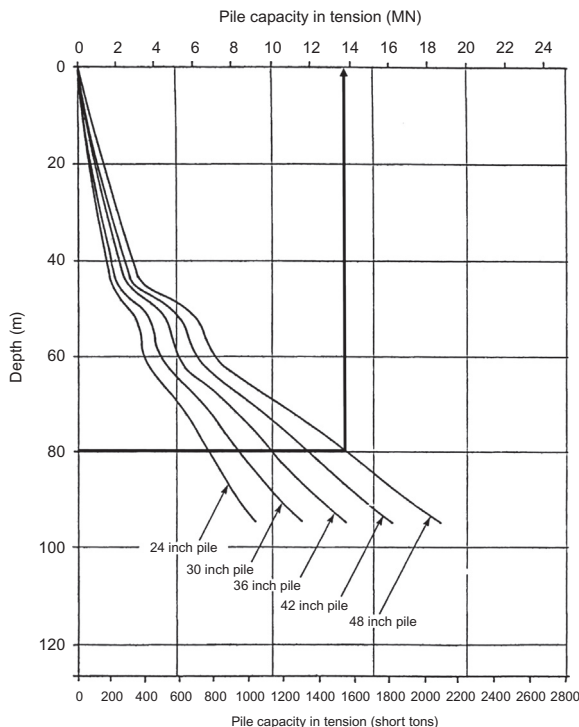


Figure 6.19 Relationship between the pile capacity in tension and different pile diameters.

should be overflowed to ensure that the initial mixture of cement and seawater is cleared. Pressures should be carefully controlled to prevent forcing the grout out from under the jacket sleeve; usually, this exit is restricted by a grout retainer, but many times the grout retainer will have been damaged during pile driving. Therefore, a second entry grouting pipe is often provided, to permit the first grout to set and form a plug; then the main grouting is carried out through the upper entry port.

6.10.1.1 Computation of allowable axial force

Until now, little research has been done into the transfer of axial force from the pile to the leg through the grouting. The transfer of axial force from the pile to the leg is accomplished by the bond strength between the pile and the leg through the grouting. The allowable axial load transfer should be calculated as the smaller value from pile or leg of the force, calculated by a multiplication of the contact area between the grout and steel surfaces, and the allowable bond strength due to axial load f_{ba} . The value of the allowable bond strength due to axial load, f_{ba} , should be taken as 0.138 MPa for API loading conditions 1 and 2 and 26.7 psi (0.184 MPa) for loading conditions 3 and 4, where the API loading conditions are

1. $E_o + DL + L_{\max}$
2. $E_o + DL + L_{\min}$

3. $ED + DL + L_{\max}$

4. $ED + DL + L_{\min}$

where E_o is the operating environmental condition; DL is the dead load; $L_{\max}$ and $L_{\min}$ are the maximum and minimum live loads, respectively, and in both cases, the live load is appropriate to the normal operating condition of the platform; ED is the design environmental condition in extreme environmental conditions.

In the case of higher values of axial load, shear keys are required at the interface between the steel and the grout, and the value of the nominal allowable bond strength for axial load transfer f_{ba} , for loading conditions 1 and 2 should be taken as

$$f_{ba} = 0.138 + 0.5f_{cu} \cdot \frac{h}{s}, \text{ MPa} \quad (6.31)$$

For loading conditions 3 and 4, f_{ba} should be taken as

$$f_{ba} = 0.184 + 0.67f_{cu} \cdot \frac{h}{s}, \text{ MPa} \quad (6.32)$$

where f_{cu} = unconfined grout compressive strength (in MPa), h = shear key outer dimension (in mm), and s = shear key spacing (in mm). Shear keys designed according to [equations \(6.31\) and \(6.32\)](#) should be detailed in accordance with the following requirements:

1. Shear keys may be circular hoops at spacing s or a continuous helix with a pitch of S_s , as in [Figure 6.20](#).
2. Shear keys should be one of the types indicated in [Figure 6.21](#).
3. For driven piles, shear keys on the pile should be applied to sufficient length to ensure that, after driving, the length of the pile in contact with the grout has the required number of shear keys.
4. Each shear key cross section and weld should be designed to transmit the part of the connection capacity that is attributable to the shear key for loading conditions 1 and 2.

The shear key and weld should be designed at basic allowable steel and weld stresses to transmit an average force equal to the shear key bearing area multiplied by $1.7 f_{cu}$, except for a distance of two pile diameters from the top and the bottom end of the connections, where $2.5 f_{cu}$ should be used. The following limitations should be observed when designing a connection:

- $17.25 \text{ MPa} \leq f_{cu} \leq 110 \text{ MPa}$
- Sleeve geometry, $D_s/t_s \leq 80$
- Pile geometry, $D_p/t_p \leq 40$
- Grout annulus geometry, $7 \leq D_g/t_g \leq 45$
- Shear key spacing ratio, $2.5 \leq D_p/s \leq 8$
- Shear key ratio, $h/s \leq 0.10$
- Shear key shape factor, $1.5 \leq w/h \leq 3$
- $f_{cu} (h/s) \leq 5.5 \text{ MPa}$.

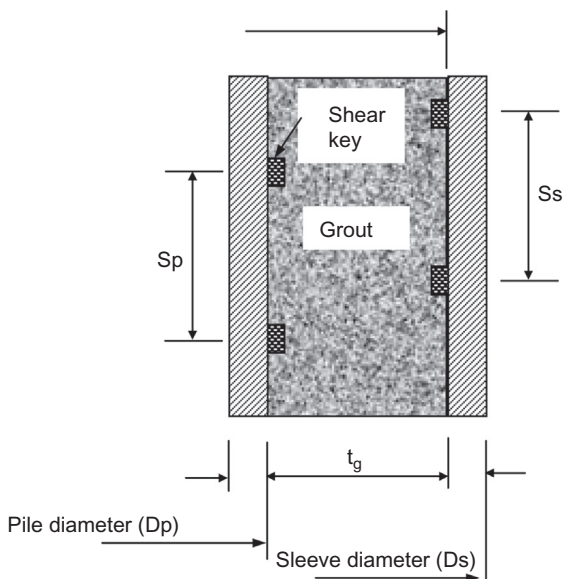


Figure 6.20 Grouting the annulus between piles and legs.

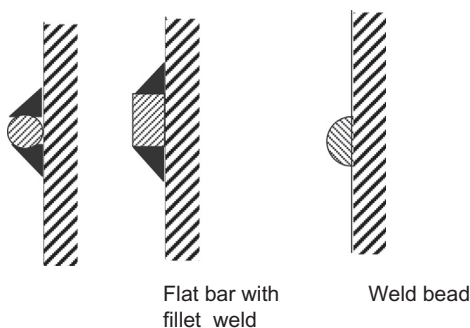


Figure 6.21 Recommended API shear key detail.

Example 6.7

Calculate the composite pipe pile of Figure 6.22, where the actual moment of inertia is described as follows:

$$D_{o1} = 1800 \text{ mm}$$

$$D_{i1} = 1730 \text{ mm}$$

$$D_{o2} = 1600 \text{ mm}$$

$$D_{i2} = 1540 \text{ mm}$$

$$I_1 = \pi[(D_{o1})^4 - (D_{i1})^4]/64 = 7,5601,916,209 \text{ mm}^4$$

$$I_2 = 45,607,749,335$$

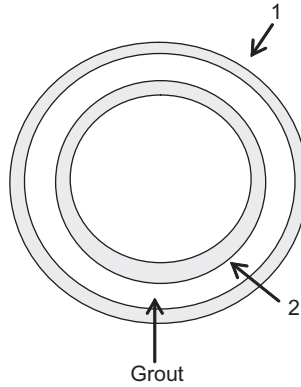


Figure 6.22 Composite pipe pile of [Example 6.7](#).

$$\begin{aligned}
 I_t &= I_1 + I_2 = 1.2121 \times 10^{11} \text{ mm}^4 \\
 A_1 &= \pi[(D_{o1})^2 - (D_{i1})^2]/4 = 194,071.8862 \text{ mm}^2 \\
 A_2 &= 147,969.014 \text{ mm}^2 \\
 A_G &= 339,998.8649 \text{ mm}^2
 \end{aligned}$$

The equivalent composite pile is

$$\begin{aligned}
 I &= 1.2121 \times 10^{11} \text{ mm}^4 \\
 D_o &= 1800 \text{ mm} \\
 D_i &= \{[I(64)/\pi] - (D_o)^4\}^{0.25} = 1683.28 \text{ mm} \\
 t &= D_o - D_i = 58.36 \text{ mm} \\
 \text{Area} &= \pi[(D_{o1})^2 - (D_{i1})^2]/4 = 319,317.75 \text{ mm}^2 \\
 \text{Elastic modulus for steel, } E_s &= 210,000 \text{ MPa} \\
 \text{Elastic modulus for grout, } E_G &= 30,000 \text{ MPa} \\
 E_t &= \{E_s [(D_{o1})^2 - (D_{i1})^2] + E_s (D_{o2})^2 - (D_{i2})^2\} + E_G [(D_{i1})^2 - (D_{o2})^2]/D_o = 32,235 \text{ MPa}
 \end{aligned}$$

6.11 Mud mat design

The geotechnical investigation report should include the bearing capacity analysis at the mudline. In most cases, it will be done for circular, triangular, square, and rectangular mud mats (with ratios $L/B = 1.5$ and $L/B = 2$) and define the where the lengths of the sides range. In addition, the pure vertical bearing capacity analysis for mud mats with skirt length of 1 m and 2 m also are performed by considering the skirt depth as the foundation base level.

Temporary on-bottom stability of the platform jacket following touchdown on the seabed and prior to piling is provided by mud mats. The ultimate bearing capacity of a mud mat is calculated as

$$Q_u = q_u A$$

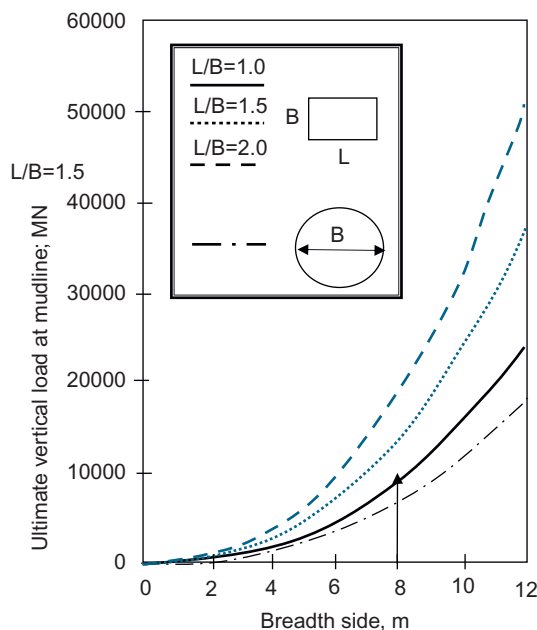


Figure 6.23 Relationship between foundation size and ultimate strength.

where

Q_u = ultimate bearing capacity

q_u = ultimate unit bearing capacity

A = mud mat area

ISO 19901-4(2003) provides a further analysis of bearing capacity for combined loading performed according to mud mat size. Figure 6.23 presents 10.0×10.0 m, 10.0×7.6 m, and 8.0×12.0 m mud mats equipped with 0 m, 1 m, and 2 m long skirts. For mud mats without skirts, full contact between mud mats and seabed has been assumed.

The ultimate bearing capacities discussed in the examples do not include a factor of safety. A factor of 2.0 is recommended by API RP2A-WSD (2000) for pure vertical loading.

The relation between the vertical load on the mud mat and the horizontal load and depend on whether there is a skirt pile or not and the depth of the skirt pile. Figures 6.24 through 6.26 present cases of no skirt pile and skirts with a depth 1 and 2 m.

Example 6.8

Based on Figure 6.23, if the mud mat is square with area $8 \text{ m} \times 8 \text{ m}$, what will be the maximum vertical load?

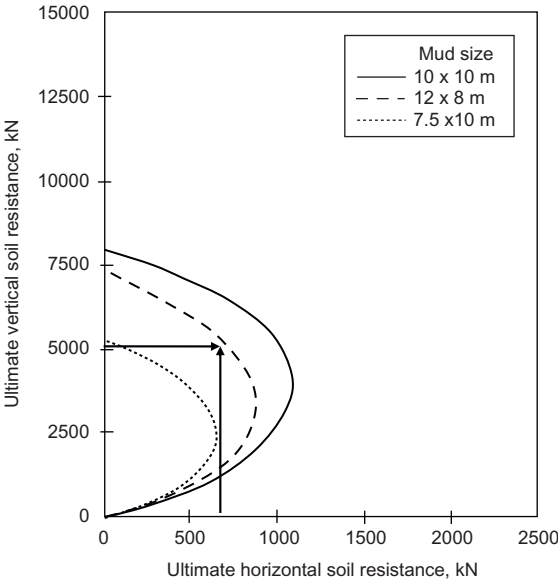


Figure 6.24 Relationship between ultimate vertical and horizontal soil resistance with no skirt.

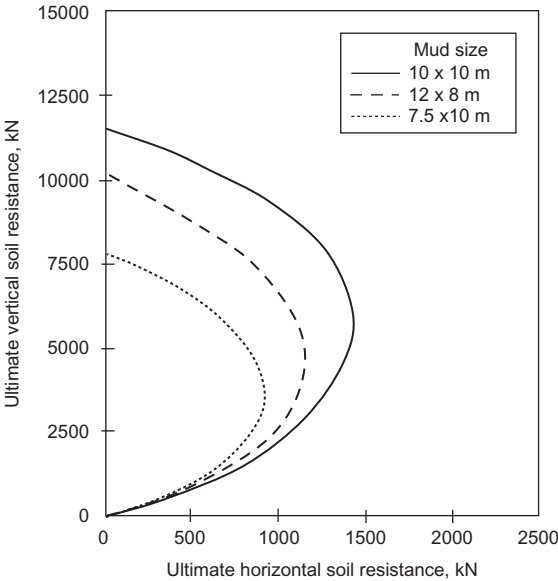


Figure 6.25 Relationship between ultimate vertical and horizontal soil resistance with skirt 1 m.

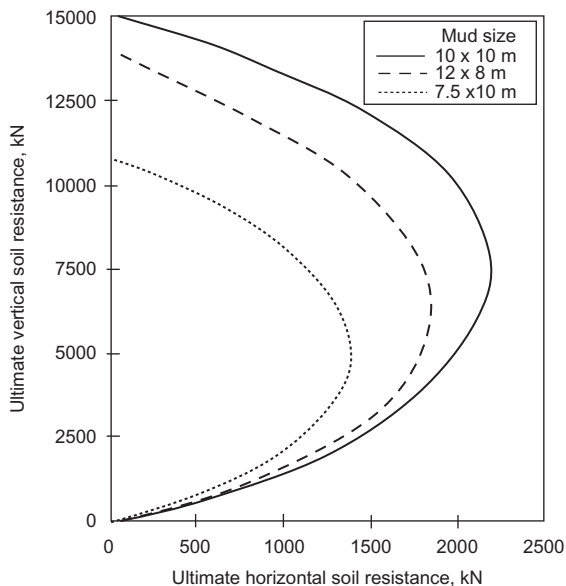


Figure 6.26 Relationship between ultimate vertical and horizontal soil resistance with skirt 2 m.

Solution As in Figure 6.23, the maximum ultimate strength is 9000 kN, around 900 ton

Example 6.9

Select the better dimensions of a mud mat with no skirt:

1. The vertical load is 5000 kN and the horizontal load is 1000 kN.
2. The vertical load 5000 kN and the horizontal load is 600 kN.

Solution As per Figure 6.24,

1. In this case, 10×10 m is the better solution.
2. In this case, 8×12 m is the better solution.

Example 6.10

Choose the better dimensions of a mud mat with a skirt of 1 m:

1. The vertical load is 5000 kN and the horizontal load is 1000 kN.
2. The vertical load 5000 kN and the horizontal load is 600 kN.

Solution As per Figure 6.25,

1. In this case, 8.0×12 m is the better solution.
2. In this case, 7.6×12 m is the better solution.

Further reading

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- American Society for Testing and Materials, 2004. Standard Method of Deep Quasi-Static Cone and Friction-Cone Penetration Tests of Soil. ASTM International, West Conshohocken, PA, ASTM Standard D 3441.
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